Optimal Market-Making with Hawkes Process: A Markovian Approximation Approach via Mercer's Expansion

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Abstract

We study an optimal market-making problem under which the order flow and liquidity level are driven by self-exciting Hawkes processes. To overcome the challenge brought by the non-Markovian structure of the problem, we propose a Markovian lifting approach where the Hawkes kernels are approximated by their truncated Mercer's expansions. This enables dynamic programming and in turn computationally feasible procedures to solve the market-making problem. The theoretical convergence of the approximated solution to the true value function is proven. Our numerical findings show that ignoring persistence with the order flow and liquidity level underestimates adverse selection risk, whereas explicitly modelling them improves the robustness and profitability of the market-making strategies.

Keywords: Market-making, Hawkes process, algorithmic trading, kernel decomposition, HJB equation, deep learning, physics-informed neural network.

1 Introduction

The recent technological and infrastructural developments in electronic trading have led to an explosive growth in high frequency trading (HFT) activities. European Central Bank reported in 2020 that HFT accounts for about 50% of trading volume in the US equity markets.[1] While HFT encompasses a broad variety of strategies, majority of the HFT flow can be attributed to market-making (see for example Hagströmer and Nordén (2013) and Menkveld (2013)). In a nutshell, a market-maker engages in continuous liquidity provision via simultaneously posting bid and offer prices on a risky asset. Their goal is to design the optimal dynamic quoting strategy to maximise profit from the spreads collected while mitigating inventory risk.

The theoretical formulation of such dynamic optimisation problem has spawned a large literature on the mathematical models of algorithmic market-making. The seminal works of Ho and Stoll (1981) and Avellaneda and Stoikov (2008) popularise continuous-time stochastic control of jump processes as a reduced-form approach to market-making. In such frameworks, cumulative long and short inventory level are modelled as controlled pure-jump processes. The market-maker can dynamically adjust bid and ask price on a risky asset where the instantaneous filled probabilities (i.e. the jump rates of inventory processes) are typically some parametric functions of the posted depths. The market-maker then solves an optimal control problem by finding the best choice of the spread processes to maximise the expected total profit-and-loss (or its risk functional) penalised by the inventory holdings. The excellent textbooks of

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[1]Research Bulletin No. 78 (15 December 2020), European Central Bank. Retrieved from `ecb.europa.eu`.

Cartea et al. (2015) and Guéant (2016) offer a comprehensive overview of the baseline models. Meanwhile, a number of extensions have been investigated in the extant literature. Examples of further modelling consideration include alpha signal (Cartea and Wang (2020)), position limits (Guéant et al. (2013)), model misspecification (Cartea et al. (2017)), feedback effect of order flow on arrival rate and shape of the limit order book (Cartea et al. (2018)), multi-asset market-making (Bergault et al. (2021)), alternative underlying instruments like options (Stoikov and Sağlam (2009)), competing market-makers (Boyce et al. (2025)), among others.

Empirically, market order flows exhibit clustering phenomenon with strong self- and cross-excitation behaviours. Several studies (e.g. Large (2007), Hardiman et al. (2013), Bacry et al. (2015), Lallouache and Challet (2016), to name a few) show that Hawkes process can satisfactorily capture and reproduce important microstructural stylised facts of market order flows. These observations motivate the use of Hawkes process in some contemporary works on algorithmic market-making. Lai et al. (2023) develop a deep-hedging approach for stock option market-making under Hawkes process. Kumar (2024) considers a neural Hawkes process where optimal market-making strategies are estimated under a simulated agent-based framework. Jain et al. (2025) study an impulse control formulation of market-making under exponential Hawkes kernel and solve the problem by reinforcement learning.

If market order flow follows a self-exciting process, then one might think that the arrival of a market order will increase the chance of execution of a market-maker's limit order (while holding the quoted spread unchanged) at least over a short horizon in the future due to the spike in the order arrival intensity. This however is not necessarily true because the *conditional* filled probability of a limit order (i.e. the probability of successful limit order execution conditional on the arrival of a directionally-matched market order) in general depends on both the quoted spread as well as an array of "liquidity" factors, and the latter could change dynamically in response to the arrival of the market orders. In this paper, we broadly interpret "liquidity" as the external market characteristics which influence the conditional filled probability of a limit order. These factors could for example be related to the bid-ask spread or more generally the shape of the limit order book, the distribution of the size of the incoming market orders, limit orders traffic from the other market participants, etc.[2]

Many existing frameworks of algorithmic market-making (with Cartea et al. (2018) being one of the notable exceptions) however model the conditional filled probability as a static function of the quoted spread with constant parameters, which fail to take into the account the interaction between order flow and market liquidity. Although it is hard to quantify liquidity precisely, there is empirical evidence suggesting that market order flow can trigger change in the limit order book structure. For example, it is observed that the arrival of a market order tends to result in an increase of new limit orders submitted at the same price. See for example Bonart and Gould (2017) and Weber and Rosenow (2005). The increase in limit orders submission signifies improvement in liquidity, and consequently a market-maker should post more competitive quotes to increase the chance of execution and to avoid being undercut. Motivated by these considerations, it is of great theoretical and practical interest to address the modelling gap by extending the Hawkes process framework for market-making to incorporate the *simultaneous* dependence of order arrival intensity and liquidity level on market orders flow.

The contributions of this paper are threefold. First, we develop a market-making framework which is capable of capturing the transient spikes of both the order arrival rates and the liquidity level triggered by incoming market orders. Specifically, market buy/sell order flows are described by a pair Hawkes processes with some memory kernels. The conditional filled probability of the market-maker's posted limit buy/sell order is modelled by a function of the quoted spread, and this function is parameterised by a state variable representing the current liquidity level of the market. The liquidity variable (denoted as

[2]The microstructural foundations of the conditional filled probability in exponential form are discussed in Avellaneda and Stoikov (2008) and Cartea et al. (2018). See also Remark 2.1.

$\kappa = (\kappa(t))_{t\geq 0})$ itself is stochastic, following a Volterra-type dynamics which will jump upon the arrival of a market order but the shock will decay over time as prescribed by another memory kernel. A similar model is considered in Cartea et al. (2018), where an incoming market order will cause jumps in the market order arrival intensity as well as the process κ. The decay effect of the shocks is captured by a mean-reversion component in Cartea et al. (2018), whereas in this paper we instead consider a generic memory kernel under a full-fledged Hawkes process as motivated by its empirical suitability and flexibility to describe market order flows.

Away from exponential memory kernel, it is generally difficult to approach a stochastic control problem featuring Hawkes processes due to the lack of Markovian properties. Our second contribution is to overcome this challenge by a Markovian lift technique. The theoretical foundation of this approach is based on the Mercer's expansion, which inspires us to approximate the kernels by a finite sum of some "feature maps". Markovian lift (based on Mercer's or Bernstein's theorem) has proven to be a useful computational and theoretical tool to study financial modelling problems with Volterra-type dynamics such as transient market impact and rough volatility. See Buet-Golfouse and Martin (2023), Hamaguchi (2024), and the references therein. For our market-making problem, the feature maps extracted from the Hawkes memory kernels serve as the additional auxiliary state variables such that the dynamics of the control problem can be approximated by a Markovian system. Standard dynamic programming approach then becomes applicable again which enables us to derive an HJB equation and obtain the approximated optimal quotes in feedback form. We justify this approximation approach by rigorously showing that the sequence of the approximated value functions indeed converges to the actual value function of the problem.

The resulting HJB equation is a high-dimensional one and it cannot be conveniently solved by traditional numerical methods like finite difference. As our third contribution, we propose a numerical algorithm to estimate the optimal market-making strategy based on deep neural network. The numerical procedure consists of two steps. First, we represent the feature maps of a kernel by some neural network which parameters are then learnt via minimising the error between the truncated Mercer series and the target kernel (c.f. Joseph and Jain (2024) and Buet-Golfouse and Martin (2023)). Once the approximated kernel becomes available, in the second step we utilise a physics-informed neural network (PINN) to solve the HJB equation. The value function is parameterised by another neural network with time and the lifted states as the inputs, and then the model is trained by minimising the error associated with the residuals of the HJB equation and the terminal condition. The numerical solutions allow us to gain insights into the behaviours of the optimal quotes and the roles of order flow and liquidity persistence. In particular, we find that incorporation of self-exciting liquidity would encourage market-makers to more proactively time the widening and tightening of the quoted spreads in response to order flows. Our simulation studies further reveal the importance of incorporating order flow and liquidity persistence in form of a Hawkes process, where a naive market-maker who fails to take these into the account would suffer from a significant decline in their trading performance.

The rest of the paper is organised as follows. Section 2 provides a mathematical description of the underlying market-making framework and outlines a Markovian approximation approach to simplify the problem. A theoretical convergence result is also given to justify our approximation. Then the HJB equation associated with the approximated optimisation problem is derived in Section 3. In Section 4, we develop a physics-informed neural network to solve a high-dimensional PDE to numerically identify the optimal market-making strategies and discuss the financial implications. Section 5 concludes. Miscellaneous technical materials and proofs are collected in the appendix.

2 The market-making problem

2.1 Modelling setup

Fix $T > 0$ as the trading horizon of the market-maker. Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\leq T}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, which supports all the stochastic processes and random variables arising in our modelling framework.

Let $(M^+, M^-) = (M^+(t), M^-(t))_{t\geq 0}$ be a pair of Hawkes processes, where $M^+(t)$ and $M^-(t)$ represent respectively the cumulative number of buy and sell orders arriving at the market over the time interval $[0, t]$. The self-exciting intensity processes of $M^\pm$, denoted by $(\lambda^\pm(t))_{t\geq 0}$, satisfy

$$\lambda^\pm(t) = \lambda_0^\pm(t) + a_{\lambda^\pm} \int_0^t K^\pm(t, s) dM^\pm(s). \tag{2.1}$$

$\lambda_0^\pm(t)$ are some strictly positive and bounded deterministic processes representing the baseline intensities of the order arrival processes. $K^\pm$ is given by $K^\pm(t, s) := g^\pm(|t - s|)$, where $g^\pm : [0, T] \to \mathbb{R}_+$ is a pair of continuous decreasing functions (or kernels) which L^1 norms satisfy the stability conditions $||g^\pm||_1 := \int_0^T g^\pm(u)du < 1/a_{\lambda^\pm}$. Here, $a_{\lambda^\pm}$ are two non-negative constants capturing the instantaneous jump size of $\lambda^\pm$ when $M^\pm$ jumps, i.e. an incoming order is likely to trigger new orders of the same direction due to the elevated jump intensities. The impact of a market order on the jump intensity however decays over time due to the kernels $K^\pm$. The mid-price process of an underlying stock $S = (S(t))_{t\geq 0}$ is assumed to follow the jump-diffusion dynamics

$$dS(t) = \sigma\, dW(t) + \epsilon^+ dM^+(t) - \epsilon^- dM^-(t),$$

where $W = (W(t))_{t\geq 0}$ is a one-dimensional Brownian motion independent of $M^\pm$, $\sigma > 0$ is a volatility coefficient and $\epsilon^\pm \geq 0$ are two constants capturing the (permanent) price impact caused by each unit of the incoming order flows.

At each time point $t \in [0, T]$, the market-maker chooses spread levels $\delta^-(t) \geq 0$ and $\delta^+(t) \geq 0$ which are required to be non-negative, bounded and predictable. The choices translate into posted limit buy and sell order for one unit of stock at price level $S(t-) - \delta^-(t)$ and $S(t-) + \delta^+(t)$ respectively. Conditional on a sell (resp. buy) order arriving at the market at time t such that $M^-(t) - M^-(t-) = 1$ (resp. $M^+(t) - M^+(t-) = 1$), the market-maker's posted limit buy (resp. sell) order will be filled with probability $p^-(t-)$ (resp. $p^+(t-)$), where

$$p^\pm(t) := \exp\left(-\kappa^\pm(t)\delta^\pm(t)\right) \in (0, 1]$$

for some pair of positive processes $(\kappa^+(t))_{t\geq 0}$ and $(\kappa^-(t))_{t\geq 0}$ that we will soon specify. Exponential form of (conditional) filled probability has been widely considered in the literature - see for example Avellaneda and Stoikov (2008), Cartea et al. (2018), Guéant et al. (2013), among others. Generally, the quantity $\kappa^\pm(t)$ can be interpreted as an elasticity measure at time t. Larger the value $\kappa^\pm$, more sensitive the filled probability is with respect to the widening of the quoted spread $\delta^\pm$. In this paper, we will refer to $\kappa^\pm(t)$ as the *market liquidity level*. In a highly liquid market, the filled probability is very sensitive to quoted spreads due to various reasons such as tight bid-ask spread of the limit order book, aggressive trading from the competing liquidity providers, etc. Specifically, the values of $\kappa^\pm(t)$ shall be large in a very liquid market. The individual quantities of $\kappa^+(t)$ and $\kappa^-(t)$ further distinguish the demand and supply side of the market liquidity.

Remark 2.1. Avellaneda and Stoikov (2008) show that if the size of the incoming market orders has a power law density and the shape of the limit order book is such that the price depth is proportional to the logarithm of the cumulative volume on the book, then the conditional filled probability under a quoted

spread δ is proportional to $\exp(-\kappa\delta)$ where κ depends on the exponent of the order flow's power law distribution and a proportional constant governing the shape of the limit order book. In our reduced-form approach, we abstract the precise sources of the (stochastic) κ and simply refer to it as a liquidity level proxy.

Many standard market-making models assume the conditional filled probabilities are static functions of the quoted spreads only, which imply the liquidity level is constant. To capture the empirical observations that an incoming market order flow tends to trigger a temporary increase in the market liquidity, we postulate that the liquidity measures $\kappa^{\pm}(t)$ have a mutually-exciting dependence on the market order flows such that

$$\kappa^{\pm}(t) = \kappa_0^{\pm}(t) + a_{\kappa^{\pm}} \int_0^t K_{\kappa}^{\pm}(t,s) dM_s^{\pm}. \tag{2.2}$$

In the above, $\kappa_0^{\pm}(t)$ are some strictly bounded positive deterministic processes corresponding to the baseline liquidity levels, $a_{\kappa^{\pm}} \geq 0$ are constant representing the liquidity shocks triggered by an incoming market order and $K_{\kappa}^{\pm}(t,s) := g_{\kappa}^{\pm}(|t-s|)$ with $g_{\kappa}^{\pm} : [0,T] \to \mathbb{R}_+$ being some continuous and decreasing functions. Note that there is no need to impose any constraint on the L^1 norms of $g_{\kappa}^{\pm}$.

Remark 2.2. Cartea et al. (2018) consider stochastic jump-intensities and liquidity levels that exhibit mean-reversion but they do not use memory kernels with decay. In contrast, we consider a full-fledged Hawkes process with more generic kernels.

The net inventory process of the market-maker is denoted by $Q = (Q(t))_{t\geq 0}$ with some initial level $Q(0-) = q_0$. Introduce two constant integers $\underline{q} < 0 < \bar{q}$ representing the position limits such that the market-maker can no longer place a limit buy (resp. sell) order when the net inventory level reaches $\bar{q}$ (resp. $\underline{q}$). One can write

$$Q(t) = q_0 + N^-(t) - N^+(t),$$

where $N^+ = (N^+(t))_{t\geq 0}$ (resp. $N^- = (N^-(t))_{t\geq 0}$) correspond to the cumulative units of stocks sold (resp. purchased) by the market-maker over $[0,t]$. We model $N^{\pm}$ as the thinned versions of $M^{\pm}$ with $p^+(t-)\mathbb{1}_{\{Q(t-)>\underline{q}\}}$ as the thinning probability of M^+ and $p^-(t-)\mathbb{1}_{\{Q(t-)<\bar{q}\}}$ as the thinning probability of M^-. Formally, $N^{\pm}$ can be defined as

$$N^+(t) = \sum_{i=1}^{M_t^+} \mathbb{1}_{\{U_i^+ < p^+(\tau_i^+-), Q(\tau_i^+-)>\underline{q}\}}, \qquad N^-(t) = \sum_{i=1}^{M_t^-} \mathbb{1}_{\{U_i^- < p^-(\tau_i^--), Q(\tau_i^--<\bar{q})\}}, \tag{2.3}$$

where $\{\tau_i^{\pm}\}_{i\in\mathbb{N}}$ are the jump times of $M^{\pm}$ and $\{U_i^{\pm}\}_{i\in\mathbb{N}}$ are some i.i.d. $\mathcal{F}_{\tau_i^{\pm}}$-measurable uniform random variables which are independent of W and $M^{\pm}$. Consequently, when a buy order arrives at the market say at time $t = \tau_i^+$ (such that M^+ jumps), the market marker's limit sell order will be executed (i.e. N^+ jumps as well at $t = \tau_i^+$) if and only if the current inventory $Q(t-)$ has not yet reached the lower limit $\underline{q}$ and that the event $U_i^+ < p^+(t-)$ materialises where the chance of occurrence is $p^+(t-)$. Similar mechanism holds for the order filling behaviours of the market-maker's limit buy order.

We will also consider a special case that $\underline{q} = -\infty$ and $\bar{q} = +\infty$, i.e. there are no position limits imposed on the market-maker's inventory. Then $N^{\pm}$ will simply become the thinned versions of $M^{\pm}$ with thinning probability processes $(p^{\pm}(t-))_{t\geq 0}$. This specification can simplify some of our theoretical results.

Finally, the cash process of the market-maker $X = (X(t))_{t\geq 0}$ satisfies

$$dX(t) = (S(t-) + \delta^+(t))dN^+(t) - (S(t-) - \delta^-(t))dN^-(t), \qquad X(0) = x_0,$$

for some initial wealth $x_0 \in \mathbb{R}$.

The goal of the market-maker is to find the optimal spreads $(\delta^-(t), \delta^+(t))_{t\geq 0}$ to maximise the expected value of their cash proceeds penalised by the inventory. Mathematically, they solve the optimisation problem

$$V := \sup_{(\delta^-(t),\delta^+(t))_{t\geq 0}\in\mathcal{A}} \mathbb{E}\left[X(T) + Q(T)(S(T) - \alpha Q(T)) - \beta \int_0^T Q(u)du\right], \tag{2.4}$$

where $\alpha \geq 0$ and $\beta \geq 0$ are the penalty parameters on the terminal and running inventory respectively. The admissible set $\mathcal{A}$ is defined as

$$\mathcal{A} := \left\{(\delta^-(t), \delta^+(t))_{t\geq 0} : (\delta^-(t), \delta^+(t))_{t\geq 0} \text{ is predictable, } \delta_t^{\pm} \in [0, C] \text{ for all } t \in [0, T]\right\}, \tag{2.5}$$

where $C > 0$ is some constant representing the upper bound of the spreads..

2.2 A Markovian approximation of the problem

It is theoretically and numerically challenging to solve (2.4) directly because the underlying dynamics are not Markovian due to the presence of the kernels $K^{\pm}(\cdot)$ and $K_\kappa^{\pm}(\cdot)$. In order to proceed, we will approximate the original state dynamics by a Markovian system. The theoretical foundation is the Mercer's theorem which is stated below.

Theorem 2.3 (Mercer (1909))**.** *Let $F : [0, T] \times [0, T] \to \mathbb{R}$ be a continuous, symmetric, and positive semidefinite function. Define the integral operator $\mathcal{G} : L^2([0, T]) \to L^2([0, T])$ by*

$$(\mathcal{G}f)(s) := \int_0^T F(t, s)f(t)\, dt. \tag{2.6}$$

Then there exists an orthonormal basis $\{\varphi_n\}_{n=1}^{\infty}$ of $L^2([0, T])$ consisting of eigenfunctions of $\mathcal{G}$, with corresponding non-negative eigenvalues $\{\xi_n\}_{n=1}^{\infty}$, such that

$$F(t, s) = \sum_{n=1}^{\infty} \xi_n\, \varphi_n(t)\, \varphi_n(s), \qquad s, t \in [0, T], \tag{2.7}$$

where the series converges absolutely and uniformly on $[0, T] \times [0, T]$.

Suppose the kernels in the order arrival intensity processes (2.1) are such that $K(t, s) := g^{\pm}(|t - s|)$ satisfy the required conditions in Theorem 2.3, then the theorem suggests that they can be reasonably approximated by a finite sum of some "feature maps" $\phi_n^{\pm}(t)$. Specifically, for some sufficiently large integer n, we expect

$$K^{\pm}(t, s) = \sum_{j=1}^{\infty} \xi_j^{\pm}\varphi_j^{\pm}(t)\varphi_j^{\pm}(s) \approx \sum_{j=1}^{n} \xi_j^{\pm}\varphi_j^{\pm}(t)\varphi_j^{\pm}(s) = \boldsymbol{\phi}_\lambda^{\pm,\top}(t)\boldsymbol{\phi}_\lambda^{\pm}(s) =: K_n^{\pm}(t, s), \tag{2.8}$$

where $\boldsymbol{\phi}_\lambda^{\pm} : [0, T] \to \mathbb{R}^n$ are defined via $\boldsymbol{\phi}_\lambda^{\pm}(t) := [\phi_1^{\pm}(t), \phi_2^{\pm}(t), ..., \phi_n^{\pm}(t)]^\top$ with $\phi_j^{\pm}(t) := \sqrt{\xi_j^{\pm}}\varphi_j^{\pm}(t)$. Likewise, the $K_\kappa^{\pm}$ in the liquidity level processes (2.2) can also be approximated as

$$K_\kappa^{\pm}(t, s) = \sum_{j=1}^{\infty} \xi_{\kappa,j}^{\pm}\varphi_{\kappa,j}^{\pm}(t)\varphi_{\kappa,j}^{\pm}(s) \approx \sum_{j=1}^{n} \xi_{\kappa,j}^{\pm}\varphi_{\kappa,j}^{\pm}(t)\varphi_{\kappa,j}^{\pm}(s) = \boldsymbol{\phi}_\kappa^{\pm,\top}(t)\boldsymbol{\phi}_\kappa^{\pm}(s) =: K_{\kappa,n}^{\pm}(t, s), \tag{2.9}$$

for some $\boldsymbol{\phi}_\kappa^{\pm} : [0, T] \to \mathbb{R}^n$, provided that $K_\kappa^{\pm}(t, s) = g_\kappa^{\pm}(|t - s|)$ satisfy the conditions in Theorem 2.3.

With the approximations of the kernels via the Mercer's expansion, we can now represent the intensity of the incoming market orders as

$$\begin{aligned}
\lambda^{\pm}(t) &\approx \lambda_0^{\pm}(t) + a_{\lambda^{\pm}} \int_0^t \boldsymbol{\phi}_\lambda^{\pm,\top}(t)\boldsymbol{\phi}_\lambda^{\pm}(s)dM^{\pm}(s) \\
&= \lambda_0^{\pm}(t) + a_{\lambda^{\pm}}\boldsymbol{\phi}_\lambda^{\pm,\top}(t) \int_0^t \boldsymbol{\phi}_\lambda^{\pm}(s)dM^{\pm}(s) \\
&= \lambda_0^{\pm}(t) + a_{\lambda^{\pm}}\boldsymbol{\phi}_\lambda^{\pm,\top}(t)\boldsymbol{\Lambda}_\lambda^{\pm}(t),
\end{aligned}$$

where $\mathbf{\Lambda}^{\pm}_{\lambda}(t) := \int_0^t \boldsymbol{\phi}^{\pm}_{\lambda}(s) dM^{\pm}(s) \in \mathbb{R}^n$ and $\mathbf{\Lambda}^{\pm}_{\lambda}(0) = \mathbf{0}$. We can express $\lambda^{\pm}(t)$ in differential form as

$$d\lambda^{\pm}(t) = \left[\partial_t \lambda^{\pm}_0(t) + a_{\lambda^{\pm}}(\partial_t \boldsymbol{\phi}^{\pm,\top}_{\lambda}(t))\mathbf{\Lambda}^{\pm}_{\lambda}(t)\right] dt + a_{\lambda^{\pm}} \boldsymbol{\phi}^{\pm,\top}_{\lambda}(t)\boldsymbol{\phi}^{\pm}_{\lambda}(t) dM^{\pm}(t).$$

Similarly, we have

$$\begin{aligned}\kappa^{\pm}(t) &= \kappa_0(t) + a_{\kappa^{\pm}} \int_0^t K^{\pm}_{\kappa}(s,t) dM^{\pm}(s) \\ &\approx \kappa_0(t) + a_{\kappa^{\pm}} \boldsymbol{\phi}^{\pm}_{\kappa}(t)^{\top} \mathbf{\Lambda}^{\pm}_{\kappa}(t),\end{aligned}$$

where $\mathbf{\Lambda}^{\pm}_{\kappa}(t) := \int_0^t \boldsymbol{\phi}^{\pm}_{\kappa}(s) dM^{\pm}(s) \in \mathbb{R}^n$ and $\mathbf{\Lambda}^{\pm}_{\kappa}(0) = \mathbf{0}$. We again can write $\kappa^{\pm}(t)$ in differential form as

$$d\kappa^{\pm}(t) = \left[\partial_t \kappa^{\pm}_0(t) + a_{\kappa^{\pm}}(\partial_t \boldsymbol{\phi}^{\pm,\top}_{\kappa}(t))\mathbf{\Lambda}^{\pm}_{\kappa}(t)\right] dt + a_{\kappa^{\pm}} \boldsymbol{\phi}^{\pm,\top}_{\kappa}(t)\boldsymbol{\phi}^{\pm}_{\kappa}(t) dM^{\pm}(t).$$

To summarise, the original model can now be approximated by the following Markovian system (indexed by n) with state processes $(S_n(t), X_n(t), Q_n(t), \lambda^{\pm}_n(t), \kappa^{\pm}_n(t), \mathbf{\Lambda}^{\pm}_{\lambda,n}(t), \mathbf{\Lambda}^{\pm}_{\kappa,n}(t))$ following the jump-diffusion dynamics

$$\begin{aligned}
dS_n(t) &= \sigma\, dW(t) + \epsilon^+ dM^+_n(t) - \epsilon^- dM^-_n(t), \\
dQ_n(t) &= dN^-_n(t) - dN^+_n(t), \\
&= \mathbb{1}_{\{U^-_i < p^-_n(\tau^-_{i,n}-), Q_n(\tau^-_{i,n}-) < \bar{q}\}} dM^-_n(t) - \mathbb{1}_{\{U^+_i < p^+_n(\tau^+_{i,n}-), Q_n(\tau^+_{i,n}-) > \underline{q}\}} dM^+_n(t), \\
dX_n(t) &= (S_n(t-) + \delta^+(t))\, dN^+_n(t) - (S_n(t-) - \delta^-(t)) dN^-_n(t) \\
&= (S_n(t-) + \delta^+(t)) \mathbb{1}_{\{U^+_i < p^+_n(\tau^+_{i,n}-), Q_n(\tau^+_{i,n}-) > \underline{q}\}} dM^+_n(t) \\
&\quad - (S_n(t-) - \delta^-(t)) \mathbb{1}_{\{U^-_{i,n} < p^-_n(\tau^-_{i,n}-), Q_n(\tau^-_{i,n}-) < \bar{q}\}} dM^-_n(t), \\
d\lambda^{\pm}_n(t) &= \left[\partial_t \lambda^{\pm}_0(t) + a_{\lambda^{\pm}}(\partial_t \boldsymbol{\phi}^{\pm,\top}_{\lambda,n}(t))\mathbf{\Lambda}^{\pm}_{\lambda,n}(t)\right] dt + a_{\lambda^{\pm}} \boldsymbol{\phi}^{\pm,\top}_{\lambda,n}(t)\boldsymbol{\phi}^{\pm}_{\lambda,n}(t) dM^{\pm}_n(t), \\
d\kappa^{\pm}_n(t) &= \left[\partial_t \kappa^{\pm}_0(t) + a_{\kappa^{\pm}}(\partial_t \boldsymbol{\phi}^{\pm,\top}_{\kappa,n}(t))\mathbf{\Lambda}^{\pm}_{\kappa,n}(t)\right] dt + a_{\kappa^{\pm}} \boldsymbol{\phi}^{\pm,\top}_{\kappa,n}(t)\boldsymbol{\phi}^{\pm}_{\kappa,n}(t) dM^{\pm}_n(t), \\
d\mathbf{\Lambda}^{\pm}_{\lambda,n}(t) &= \boldsymbol{\phi}^{\pm}_{\lambda,n}(t) dM^{\pm}_n(t), \\
d\mathbf{\Lambda}^{\pm}_{\kappa,n}(t) &= \boldsymbol{\phi}^{\pm}_{\kappa}(t) dM^{\pm}_n(t),
\end{aligned} \tag{2.10}$$

where W is a Brownian motion, $M^{\pm}_n$ is a pair of counting processes with stochastic jump intensities $(\lambda^{\pm}_n(t))_{t\geq 0}$, and $\{\tau^{\pm}_{i,n}\}_{i\in\mathbb{N}}$ are the jump times of $M^{\pm}_n$. The approximated value function is

$$V_n := \sup_{(\delta^-(t),\delta^+(t))_{t\geq 0}\in\mathcal{A}} \mathbb{E}\left[X_n(T) + Q_n(T)(S_n(T) - \alpha Q_n(T)) - \beta \int_0^T Q_n(u) du\right]. \tag{2.11}$$

Under suitable conditions, the approximated value function (2.11) indeed converges to the actual value function (2.4) when n becomes sufficiently large as the following theorem shows. The proof is long and technical, and is therefore deferred to the appendix.

Theorem 2.4. *Suppose the kernels $K^{\pm}(t,s) = g^{\pm}(|t-s|)$ and $K^{\pm}_{\kappa}(t,s) = g^{\pm}_{\kappa}(|t-s|)$ are such that $||g^{\pm}||_1 < a_{\lambda^{\pm}}$ and $||g^{\pm}_{\kappa}||_1 < a_{\kappa^{\pm}}$, and they satisfy the conditions in Theorem 2.3 such that they admit Mercer's expansions in form of (2.7). Recall the definition of V as the value function of the original problem (2.4) and V_n as the value function of the approximated problem (2.11) driven by the truncated kernels $K^{\pm}_n$ and $K^{\pm}_{\kappa,n}$ defined in (2.8) and (2.9). If either:*

1. *There is no position limit imposed on the inventory level, i.e. $\underline{q} = -\infty$ and $\bar{q} = +\infty$, or;*
2. *The eigenvalues $\{\xi^{\pm}_j\}_{j\in\mathbb{N}}$ of the operator $f(\cdot) \mapsto \int_0^T K^{\pm}(t,\cdot) f(t) dt$ are such that*

$$\sum_{i=1}^{\infty} \left(\sum_{j=i+1}^{\infty} (\xi^{\pm}_j)^2\right)^{\frac{1}{2}} < \infty, \tag{2.12}$$

then we have

$$\lim_{n\to\infty} |V_n - V| = 0.$$

Since $g^{\pm}$ and $g_\kappa^{\pm}$ are continuous, $K^{\pm}$ and $K_\kappa^{\pm}$ are continuous and symmetric on $[0,T]\times[0,T]$. The only additional required condition on $K^{\pm}$ and $K_\kappa^{\pm}$ for Theorem 2.3 to be applicable is the positive semidefiniteness. Examples of a parametric kernel that satisfies such condition includes Gaussian kernel $g(z) = \exp\left(-z^2/(2b^2)\right)$ for $b > 0$, the exponential (Ornstein-Uhlenbeck) kernel $g(z) = \exp(-z/a)$ for $a > 0$, and the fractional kernel $g(z) = (1 + z/a)^{-r}$ for $a > 0$ and $r > 0$, etc.

Remark 2.5. Theorem 2.3 does not cover singular kernel with $g(0+) = +\infty$ because the resulting $K(t,s) = g(|t-s|)$ would then be discontinuous along the diagonal line $t = s$. A notable example is the power-law kernel $g(z) = \frac{c}{\Gamma(\gamma)} z^{-\gamma}$ for $c > 0$ and $\gamma > 0$ where $\Gamma(\cdot)$ is the gamma function. But in practice one can approximate the power-law kernel by $g_\epsilon(z) = \frac{c}{\Gamma(\gamma)}(z+\epsilon)^{-\gamma}$ for some small $\epsilon > 0$ which is then equivalent to a fractional kernel. Furthermore, unlike the alternative Markovian lifting technique that relies on the Bernstein's theorem, our approach does not require the kernel to be completely monotone.

If one is interested in a version of the problem with inventory constraint (i.e. either $\underline{q} > -\infty$ or $\hat{q} < \infty$), then the theoretical convergence result requires an additional assumption in form of (2.12).[3] Heuristically, this condition will hold if the eigenvalues (sorted in descending order) of the operator $f(\cdot) \to \int_0^T K^{\pm}(t,\cdot) f(t) dt$ are decaying sufficiently fast. The condition is automatically satisfied by a large class of kernels. Some examples are given in the following lemma.

Lemma 2.6. *Suppress the superscript $\pm$ for brevity. Condition 2.12 is satisfied by the Gaussian kernel $g(z) = \exp\left(-z^2/(2b^2)\right)$ for any $b > 0$; the exponential (OU) kernel $g(z) = \exp(-z/a)$ for any $a > 0$; and the fractional kernel $g(z) = (1 + z/a)^{-r}$ for any $a > 0$ and $r > 0$.*

Proof. A sufficient condition for Condition 2.12 to hold is that the eigenvalues $\{\xi_n\}_n$ are decaying faster than $n^{-3/2}$, i.e. there exists some $\theta > 3/2$ and $c > 0$ such that $\xi_n \le cn^{-\theta}$ for all n.

For the Gaussian kernel, it is known that the eigenvalues exhibit decay faster than exponential and in particular $\xi_n \le ce^n/(n^n)$ for some constant c independent of n. See Example 3.6 of Azevedo and Menegatto (2014).

For the exponential kernel, it is indeed a special case of a Matérn kernel which parameterisation is $g(z) \propto (z/a)^\nu \mathcal{K}_\nu(z/a)$ for $\nu > 0$ and $a > 0$, where $\mathcal{K}_\nu(\cdot)$ is the modified Bessel function of the second kind. The eigenvalues of a Matérn kernel are known to have the order of decay given by $\xi_n = O(n^{-(2\nu+1)})$. See Section 5.2.1 of Burt et al. (2020). This specialises to $\xi_n = O(n^{-2})$ for the exponential kernel by considering $\nu = 1/2$. More generally, the result holds for a generic Matérn kernel with $\nu > 1/4$ and $a > 0$.

Similarly, the fractional kernel is the special case of a generalised Cauchy kernel, where the latter is defined by $g(z) \propto (1 + z^\rho)^{-r}$ for $r > 0$ and $\rho \in (0, 2]$. The Equation (3.13) of Lim and Teo (2009) states that the spectral density (i.e. the Fourier transform) of a generalised Cauchy kernel is asymptotically given by $S_{\rho,r}(\omega) \asymp c_{\rho,r}\omega^{-\rho-1}$ for some constant $c_{\rho,r} > 0$. Then Theorem 2 of Seeger et al. (2008) implies that the eigenvalues of the kernel asymptotically satisfy $\xi_n \le cS_{\rho,r}(c'n) = c'_{\rho,r} n^{-\rho-1}$ for large n where $c, c', c_{\rho,r} > 0$ are some other constants. We therefore deduce $\xi_n = O(n^{-2})$ for the fractional kernel which corresponds to the case of $\rho = 1$, and indeed the result can be extended to a generalised Cauchy kernel with $\rho \in (1/2, 2]$ and $r > 0$. □

[3] Note that the assumption is imposed on $K^{\pm}$ only but not on $K_\kappa^{\pm}$.

3 Derivation of the HJB equation under Markovian approximation

Theorem 2.4 provides a theoretical rationale that Problem (2.11) formulated under the Markovian dynamics outlined in Section 2.2 is a reasonable approximation of the original Problem (2.4), provided that n the number of features used in the construction of the approximated kernels $K^{\pm}$ and $K^{\pm}_{\lambda}$ is sufficiently large. In this section, we tackle Problem (2.11) by the classical dynamic programming approach.

From now on, we will suppress the subscript n in the approximated dynamics. With an abuse of notation, we will use

$$\begin{aligned} z &:= (x, q, \lambda^+, \lambda^-, \kappa^+, \kappa^-, \mathbf{\Lambda}^+_\lambda, \mathbf{\Lambda}^-_\lambda, \mathbf{\Lambda}^+_\kappa, \mathbf{\Lambda}^-_\kappa) \\ &\in \mathbb{R} \times \{\underline{q}, \underline{q}+1, ..., \bar{q}-1, \bar{q}\} \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^N =: \mathcal{Z} \end{aligned}$$

to denote the state values (excluding current stock price) of the system, and likewise we write

$$Z(t) := (X(t), Q(t), \lambda^+(t), \lambda^-(t), \kappa^+(t), \kappa^-(t), \mathbf{\Lambda}^+_\lambda(t), \mathbf{\Lambda}^-_\lambda(t), \mathbf{\Lambda}^+_\kappa(t), \mathbf{\Lambda}^-_\kappa(t)).$$

The time-t value function associated with the optimisation problem (2.11) is

$$H(t,s,z) := \sup_{(\delta^-(u),\delta^+(u))_{u\geq t}\in\mathcal{A}} \mathbb{E}\left[X(T) + Q(T)(S(T) - \alpha Q(T)) - \beta \int_t^T Q(u)du \middle| S(t-) = s, Z(t-) = z \right]$$

for $(t,s,z) \in [0,T] \times \mathbb{R} \times \mathcal{Z}$. The terminal condition of H at $t = T$ is

$$H(T,s,z = (x,q,\lambda^+,\lambda^-,\kappa^+,\kappa^-,\mathbf{\Lambda}^+_\lambda,\mathbf{\Lambda}^-_\lambda,\mathbf{\Lambda}^+_\kappa,\mathbf{\Lambda}^-_\kappa)) = x + qs - \alpha q^2.$$

The infinitesimal generator of (S, Z) is given by

$$\begin{aligned} \mathcal{L}^{\delta^+,\delta^-} H &= \frac{1}{2}\sigma^2 \partial^2_{ss} H \\ &+ \left\{ \partial_t \lambda^+_0(t) + a_{\lambda^+} \partial_t \left(\boldsymbol{\phi}^{+,\top}_\lambda(t) \right) \mathbf{\Lambda}^+_\lambda(t) \right\} \partial_{\lambda^+} H + \left\{ \partial_t \lambda^-_0(t) + a_{\lambda^-} \partial_t \left(\boldsymbol{\phi}^{-,\top}_\lambda(t) \right) \mathbf{\Lambda}^-_\lambda(t) \right\} \partial_{\lambda^-} H \\ &+ \left\{ \partial_t \kappa^+_0(t) + a_{\kappa^+} \partial_t \left(\boldsymbol{\phi}^{+,\top}_\kappa(t) \right) \mathbf{\Lambda}^+_\kappa(t) \right\} \partial_{\kappa^+} H + \left\{ \partial_t \kappa^-_0(t) + a_{\kappa^-} \partial_t \left(\boldsymbol{\phi}^{-,\top}_\kappa(t) \right) \mathbf{\Lambda}^-_\kappa(t) \right\} \partial_{\kappa^-} H \\ &+ \lambda^+ \int_0^1 \Big[H\big(t, s+\epsilon^+, t, x + (s+\delta^+)\mathbb{1}_{\{u<p^+(t), q>\underline{q}\}}, q - \mathbb{1}_{\{u<p^+(t),q>\underline{q}\}}, \\ &\qquad \lambda^+ + a_{\lambda^+} \boldsymbol{\phi}^{+,\top}_\lambda(t) \boldsymbol{\phi}^+_\lambda(t), \lambda^-, \kappa^+ + a_{\kappa^+} \boldsymbol{\phi}^{+,\top}_\kappa(t) \boldsymbol{\phi}^+_\kappa(t), \kappa^-, \\ &\qquad \mathbf{\Lambda}^+_\lambda + \boldsymbol{\phi}^+_\lambda(t), \mathbf{\Lambda}^-_\lambda, \mathbf{\Lambda}^+_\kappa + \boldsymbol{\kappa}^+_\lambda(t), \mathbf{\Lambda}^-_\kappa \big) - H(t,s,z) \Big] du \\ &+ \lambda^- \int_0^1 \Big[H\big(t, s-\epsilon^-, t, x - (s-\delta^-)\mathbb{1}_{\{u<p^-(t), q<\bar{q}\}}, q + \mathbb{1}_{\{u<p^-(t),q<\bar{q}\}}, \\ &\qquad \lambda^+, \lambda^- + a_{\lambda^-} \boldsymbol{\phi}^{-,\top}_\lambda(t) \boldsymbol{\phi}^-_\lambda(t), \kappa^+, \kappa^- + a_{\kappa^-} \boldsymbol{\phi}^{-,\top}_\kappa(t) \boldsymbol{\phi}^-_\kappa(t), \\ &\qquad \mathbf{\Lambda}^+_\lambda, \mathbf{\Lambda}^-_\lambda + \boldsymbol{\phi}^-_\lambda(t), \mathbf{\Lambda}^+_\kappa, \mathbf{\Lambda}^-_\kappa + \boldsymbol{\kappa}^-_\lambda(t) \big) - H(t,s,z) \Big] du, \end{aligned}$$

or equivalently

$$\begin{aligned}\mathcal{L}^{\delta^+,\delta^-} H = &\frac{1}{2}\sigma^2\,\partial^2_{ss}H \\ &+ \left\{\partial_t\lambda^+_0(t) + a_{\lambda^+}\,\partial_t\left(\boldsymbol{\phi}^{+,\top}_\lambda(t)\right)\,\boldsymbol{\Lambda}^+_\lambda(t)\right\}\partial_{\lambda^+}H + \left\{\partial_t\lambda^-_0(t) + a_{\lambda^-}\,\partial_t\left(\boldsymbol{\phi}^{-,\top}_\lambda(t)\right)\,\boldsymbol{\Lambda}^-_\lambda(t)\right\}\partial_{\lambda^-}H \\ &+ \left\{\partial_t\kappa^+_0(t) + a_{\kappa^+}\,\partial_t\left(\boldsymbol{\phi}^{+,\top}_\kappa(t)\right)\,\boldsymbol{\Lambda}^+_\kappa(t)\right\}\partial_{\kappa^+}H + \left\{\partial_t\kappa^-_0(t) + a_{\kappa^-}\,\partial_t\left(\boldsymbol{\phi}^{-,\top}_\kappa(t)\right)\,\boldsymbol{\Lambda}^-_\kappa(t)\right\}\partial_{\kappa^-}H \\ &+ \lambda^+\left\{e^{-\kappa^+\delta^+}\,\mathbb{1}_{\{q>\underline{q}\}}\left[H(t,s+\epsilon^+,z^+) - H(t,s+\epsilon^+,\tilde{z}^+)\right] + H(t,s+\epsilon^+,\tilde{z}^+) - H(t,s,z)\right\} \\ &+ \lambda^-\left\{e^{-\kappa^-\delta^-}\,\mathbb{1}_{\{q<\bar{q}\}}\left[H(t,s-\epsilon^-,z^-) - H(t,s-\epsilon^-,\tilde{z}^-)\right] + H(t,s-\epsilon^-,\tilde{z}^-) - H(t,s,z)\right\},\end{aligned} \tag{3.1}$$

where in (3.1) we have introduced the shorthands

$$\begin{aligned}z^+ = \Big(&x + (S_{t_-} + \delta^+),\ q-1,\ S_{t_-} + \epsilon^+,\ \lambda^+ + a_{\lambda^+}\boldsymbol{\phi}^{+,\top}_\lambda\boldsymbol{\phi}^+_\lambda,\ \lambda^-,\ \kappa^+ + a_{\kappa^+}\boldsymbol{\phi}^{+,\top}_\kappa\boldsymbol{\phi}^+_\kappa,\ \kappa^-, \\ &\boldsymbol{\Lambda}^+_\lambda + \boldsymbol{\phi}^+_\lambda,\ \boldsymbol{\Lambda}^-_\lambda,\ \boldsymbol{\Lambda}^+_\kappa + \boldsymbol{\phi}^+_\kappa,\ \boldsymbol{\Lambda}^-_\kappa\Big), \\ \tilde{z}^+ = \Big(&x,\ q,\ S + \epsilon^+,\ \lambda^+ + a_{\lambda^+}\boldsymbol{\phi}^{+,\top}_\lambda\boldsymbol{\phi}^+_\lambda,\ \lambda^-,\ \kappa^+ + a_{\kappa^+}\boldsymbol{\phi}^{+,\top}_\kappa\boldsymbol{\phi}^+_\kappa,\ \kappa^-, \\ &\boldsymbol{\Lambda}^+_\lambda + \boldsymbol{\phi}^+_\lambda,\ \boldsymbol{\Lambda}^-_\lambda,\ \boldsymbol{\Lambda}^+_\kappa + \boldsymbol{\phi}^+_\kappa,\ \boldsymbol{\Lambda}^-_\kappa\Big), \\ z^- = \Big(&x - (S_{t_-} - \delta^-),\ q+1,\ S_{t_-} - \epsilon^-,\ \lambda^+,\ \lambda^- + a_{\lambda^-}\boldsymbol{\phi}^{-,\top}_\lambda\boldsymbol{\phi}^-_\lambda,\ \kappa^+,\ \kappa^- + a_{\kappa^-}\boldsymbol{\phi}^{-,\top}_\kappa\boldsymbol{\phi}^-_\kappa, \\ &\boldsymbol{\Lambda}^+_\lambda,\ \boldsymbol{\Lambda}^-_\lambda + \boldsymbol{\phi}^-_\lambda,\ \boldsymbol{\Lambda}^+_\kappa,\ \boldsymbol{\Lambda}^-_\kappa + \boldsymbol{\phi}^-_\kappa\Big), \\ \tilde{z}^- = \Big(&x,\ q,\ S - \epsilon^-,\ \lambda^+,\ \lambda^- + a_{\lambda^-}\boldsymbol{\phi}^{-,\top}_\lambda\boldsymbol{\phi}^-_\lambda,\ \kappa^+,\ \kappa^- + a_{\kappa^-}\boldsymbol{\phi}^{-,\top}_\kappa\boldsymbol{\phi}^-_\kappa, \\ &\boldsymbol{\Lambda}^+_\lambda,\ \boldsymbol{\Lambda}^-_\lambda + \boldsymbol{\phi}^-_\lambda,\ \boldsymbol{\Lambda}^+_\kappa,\ \boldsymbol{\Lambda}^-_\kappa + \boldsymbol{\phi}^-_\kappa\Big).\end{aligned}$$

Heuristically, z reflects the current state value of the system (excluding the current stock price). When a market buy order arrives such that M^+ jumps, there are two possibilities: 1) The market-maker's limit sell order is successfully executed (which happens with probability $e^{-\kappa^+\delta^+}$ provided that $q > \underline{q}$), and then the state value will become z^+ post-jump; or 2) The market-maker's limit sell order is not filled (which will happen certainty if $q \le \underline{q}$, or with probability $1 - e^{-\kappa^+\delta^+}$ if $q > \underline{q}$), where then the new state value becomes $\tilde{z}^+$ post-jump. Similar interpretation holds for z^- and $\tilde{z}^-$. The key difference between $z^\pm$ and $\tilde{z}^\pm$ is that the values of x and q are updated in $z^\pm$ due to successful filling of the market-maker's limit orders, while those values are not updated in $\tilde{z}^\pm$ as the limit orders are not filled such that the market-maker's cash and inventory remain unchanged.

The HJB equation is now given by

$$\partial_t H - \beta q^2 + \sup_{(\delta^+,\delta^-)\in[0,C]^2} \mathcal{L}^{\delta^+,\delta^-} H = 0. \tag{3.2}$$

Consider an ansatz

$$H(t,s,z) = x + qs + h(z),$$

for some function $h(t,z)$ to be determined. The terminal condition becomes $h(T,z) = -\alpha q^2$, where we

recall that $z = (x, q, \lambda^+, \lambda^-, \kappa^+, \kappa^-, \mathbf{\Lambda}_\lambda^+, \mathbf{\Lambda}_\lambda^-, \mathbf{\Lambda}_\kappa^+, \mathbf{\Lambda}_\kappa^-)$. Upon substitution, (3.2) can now be expressed as

$$\begin{aligned}
&\partial_t h - \beta q^2 + \left\{\partial_t \lambda_0^+(t) + a_{\lambda^+}\,\partial_t\left(\boldsymbol{\phi}_\lambda^{+,\top}(t)\right)\,\mathbf{\Lambda}_\lambda^+(t)\right\}\partial_{\lambda^+} h + \left\{\partial_t \lambda_0^-(t) + a_{\lambda^-}\,\partial_t\left(\boldsymbol{\phi}_\lambda^{-,\top}(t)\right)\,\mathbf{\Lambda}_\lambda^-(t)\right\}\partial_{\lambda^-} h \\
&\quad + \left\{\partial_t \kappa_0^+(t) + a_{\kappa^+}\,\partial_t\left(\boldsymbol{\phi}_\kappa^{+,\top}(t)\right)\,\mathbf{\Lambda}_\kappa^+(t)\right\}\partial_{\kappa^+} h + \left\{\partial_t \kappa_0^-(t) + a_{\kappa^-}\,\partial_t\left(\boldsymbol{\phi}_\kappa^{-,\top}(t)\right)\,\mathbf{\Lambda}_\kappa^-(t)\right\}\partial_{\kappa^-} h \\
&\quad + \lambda^+\left\{q\epsilon^+ + h(t,\tilde{z}^+) - h(t,z)\right\} + \lambda^-\left\{-q\epsilon^- + h(t,\tilde{z}^-) - h(t,z)\right\} \\
&\quad + \lambda^+ \sup_{\delta^+\in[0,C]}\left\{e^{-\kappa^+\delta^+}\left[\delta^+ - \epsilon^+ + h(t,z^+) - h(t,\tilde{z}^+)\right]\right\}\mathbb{1}_{q>\underline{q}} \\
&\quad + \lambda^- \sup_{\delta^-\in[0,C]}\left\{e^{-\kappa^-\delta^-}\left[\delta^- - \epsilon^- + h(t,z^-) - h(t,\tilde{z}^-)\right]\right\}\mathbb{1}_{q<\bar{q}} \\
&\quad = 0.
\end{aligned} \tag{3.3}$$

Now, solving for the optimal posting depths, $\delta^{\star,+}$ and $\delta^{\star,-}$, yields the following

$$\begin{aligned}
\delta^{+,\star} = \delta^{+,\star}(t,z) &= \min\left(C, \max\left(\frac{1}{\kappa^+} + \epsilon^+ - h\left(t,z^+\right) + h\left(t,\tilde{z}^+\right), 0\right)\right),\ q > \underline{q}, \\
\delta^{-,\star} = \delta^{-,\star}(t,z) &= \min\left(C, \max\left(\frac{1}{\kappa^-} + \epsilon^- - h\left(t,z^-\right) + h\left(t,\tilde{z}^-\right), 0\right)\right),\ q < \bar{q}.
\end{aligned} \tag{3.4}$$

The HJB equation is now

$$\begin{aligned}
&\partial_t h - \beta q^2 + \left\{\partial_t \lambda_0^+(t) + a_{\lambda^+}\,\partial_t\left(\boldsymbol{\phi}_\lambda^{+,\top}(t)\right)\,\mathbf{\Lambda}_\lambda^+(t)\right\}\partial_{\lambda^+} h + \left\{\partial_t \lambda_0^-(t) + a_{\lambda^-}\,\partial_t\left(\boldsymbol{\phi}_\lambda^{-,\top}(t)\right)\,\mathbf{\Lambda}_\lambda^-(t)\right\}\partial_{\lambda^-} h \\
&\quad + \left\{\partial_t \kappa_0^+(t) + a_{\kappa^+}\,\partial_t\left(\boldsymbol{\phi}_\kappa^{+,\top}(t)\right)\,\mathbf{\Lambda}_\kappa^+(t)\right\}\partial_{\kappa^+} h + \left\{\partial_t \kappa_0^-(t) + a_{\kappa^-}\,\partial_t\left(\boldsymbol{\phi}_\kappa^{-,\top}(t)\right)\,\mathbf{\Lambda}_\kappa^-(t)\right\}\partial_{\kappa^-} h \\
&\quad + \lambda^+\left\{q\epsilon^+ + h(t,\tilde{z}^+) - h(t,z)\right\} + \lambda^-\left\{-q\epsilon^- + h(t,\tilde{z}^-) - h(t,z)\right\} \\
&\quad + \lambda^+ J(h(t,z^+) - h(t,\tilde{z}^+) - \epsilon^+, \kappa^+)\mathbb{1}_{q>\underline{q}} + \lambda^- J(h(t,z^-) - h(t,\tilde{z}^-) - \epsilon^-, \kappa^-)\mathbb{1}_{q<\bar{q}} \\
&\quad = 0,
\end{aligned} \tag{3.5}$$

where

$$J(y,\kappa) := \sup_{\delta\in[0,C]} e^{-\kappa\delta}(\delta + y) = \begin{cases} y, & y \geq \frac{1}{\kappa}; \\ \frac{e^{-1+\kappa y}}{\kappa}, & \frac{1}{\kappa} - C < y < \frac{1}{\kappa}; \\ e^{-\kappa C}(C + y), & y \leq \frac{1}{\kappa} - C. \end{cases}$$

4 Numerical solutions to the market-making problem

Equation (3.5) cannot be solved easily by standard numerical methods due to the possibly large number of state variables. In this section, we outline a procedure based on deep neural networks.

4.1 Learning the kernel feature maps

The first step is to identity the sequence of eigenfunctions (i.e. the kernel feature maps) which approximates some given memory kernel K. For some fixed integer n which represents the truncation order of the Mercer's expansion, we construct a feedforward neural network $\phi_{\boldsymbol{\theta}} : [0,T] \to \mathbb{R}^n$ of a certain architecture parameterised by $\boldsymbol{\theta}$ to approximate a target kernel $K(\cdot)$, i.e. $\boldsymbol{\theta}$ should be chosen such that

$$K(t,s) \approx \phi_{\boldsymbol{\theta}}(t)^\top \phi_{\boldsymbol{\theta}}(s).$$

The netowrk parameter $\boldsymbol{\theta}$ can be learnt by minimising the loss function

$$\mathcal{L}_{\text{kernel}}(\boldsymbol{\theta}) := \frac{1}{M_{\text{kernel}}} \sum_{i=1}^{M_{\text{kernel}}} \left\{ \left[\left(K(t_i, s_i) - \phi_{\boldsymbol{\theta}}(t_i)^\top \phi_{\boldsymbol{\theta}}(s_i) \right)^2 \right] + \left[\left(1 - ||\phi_{\boldsymbol{\theta}}(t_i)||^2 \right)^2 \right] \right\},$$

where $\{t_i, s_i\}_{i=1}^{M_{\text{kernel}}}$ are some data points uniformly and randomly sampled on $\{(t,s) \in [0,T]^2 : t \geq s\}$. The minimiser $\hat{\boldsymbol{\theta}}$ can be estimated by standard gradient descent algorithm. Since the kernels to be considered will have a reference level of one under zero time lapse, i.e., $K(t,t) = g(0) = 1$ for all $t \in [0,T]$, the second term in the loss function is introduced to penalise deviation of $\phi_{\boldsymbol{\theta}}(t)^\top \phi_{\boldsymbol{\theta}}(t)$ from one.

The above procedure is applied individually to the four kernels arising in our model: K^+, K^-, K^+_κ and K^-_κ. For simplicity, the four feedforward neural networks which approximate these kernels will share the same architecture. Denote the fitted feature maps of the kernels by $\phi(t; \hat{\boldsymbol{\theta}}_{\lambda^+})$, $\phi(t; \hat{\boldsymbol{\theta}}_{\lambda^-})$, $\phi(t; \hat{\boldsymbol{\theta}}_{\kappa^+})$ and $\phi(t; \hat{\boldsymbol{\theta}}_{\kappa^-})$ respectively.

4.2 Solving the HJB equation via Physics-informed neural network

The solution to the PDE (3.5) is now represented by another feedforward neural network $h_{\boldsymbol{\psi}} : [0,T] \times \mathcal{Z} \to \mathbb{R}$ parameterised by $\boldsymbol{\psi}$. Using the feature maps of the kernels learnt in Section 4.1, the residual of (3.5) associated with $h_{\boldsymbol{\psi}}$ can be written as

$$\begin{aligned}
\mathcal{F}[h_{\boldsymbol{\psi}}](t,z) := \partial_t h_{\boldsymbol{\psi}}(t,z) + & \sum_{\nu \in \{\lambda^+, \kappa^+, \lambda^-, \kappa^-\}} \Big(\partial_t \nu_0(t) + a_\nu \, \partial_t \boldsymbol{\phi}^\top(t; \hat{\boldsymbol{\theta}}_\nu) \boldsymbol{\Lambda}_\nu(t) \Big) \partial_\nu h_{\boldsymbol{\psi}}(t,z) \\
& + \lambda^+ \big(q\epsilon^+ + h_{\boldsymbol{\psi}}(t, \tilde{z}^+) - h_{\boldsymbol{\psi}}(t,z) \big) + \lambda^- \big(-q\epsilon^- + h_{\boldsymbol{\psi}}(t, \tilde{z}^-) - h_{\boldsymbol{\psi}}(t,z) \big) \\
& + \lambda^+ J(h_{\boldsymbol{\psi}}(t,z^+) - h_{\boldsymbol{\psi}}(t,\tilde{z}^+) - \epsilon^+, \kappa^+) \mathbb{1}_{\{q > \underline{q}\}} \\
& + \lambda^- J(h_{\boldsymbol{\psi}}(t,z^-) - h_{\boldsymbol{\psi}}(t,\tilde{z}^-) - \epsilon^-, \kappa^-) \mathbb{1}_{\{q < \bar{q}\}},
\end{aligned} \tag{4.1}$$

where we recall that $z = (x, q, \lambda^+, \lambda^-, \kappa^+, \kappa^-, \boldsymbol{\Lambda}^+_\lambda, \boldsymbol{\Lambda}^-_\lambda, \boldsymbol{\Lambda}^+_\kappa, \boldsymbol{\Lambda}^-_\kappa)$ represents all the state variables excluding the current stock price. All the partial derivatives acting on the neural networks $h_{\boldsymbol{\psi}}$ and $\phi(\cdot; \hat{\boldsymbol{\theta}}_\nu)$ can be computed by automatic differentiation.

The physics-informed neural network (PINN) approach of Raissi et al. (2019) guides us to consider a loss function which combines the interior PDE residuals and the error at the terminal time, which is given by

$$\mathcal{L}(\boldsymbol{\psi}) := \frac{1}{M_{\text{int}}} \sum_{i=1}^{M_{\text{int}}} (\mathcal{F}[h_{\boldsymbol{\psi}}](t_i, z_i))^2 + \gamma \, \frac{1}{M_{\text{term}}} \sum_{i=1}^{M_{\text{term}}} \left(h_{\boldsymbol{\psi}}(T, z_i^T) + \alpha (q_i^T)^2 \right)^2 \tag{4.2}$$

for some regularisation term $\gamma > 0$. Here, $\{t_i, z_i\}_{i=1}^{M_{\text{int}}}$ are M_{int} points randomly sampled on $[0,T) \times \mathcal{Z}$, while $\{z_i^T\}_{i=1}^{M_{\text{term}}}$ are M_{term} points randomly sampled on $\mathcal{Z}$.[4] Note that q_i^T denotes the element in z_i^T corresponding to the inventory holding. The optimal network parameters $\hat{\boldsymbol{\psi}}$ are obtained by minimising (4.2) with respect to $\boldsymbol{\psi}$. The approximated optimal quotes are now given by (3.4) upon replacing $h(\cdot)$ by the trained neural network $h_{\hat{\boldsymbol{\psi}}}(\cdot)$.

4.3 Configurations of the numerical experiments

To simplify the analysis, we assume the four different kernels in our market-making model are identical, i.e. there is a single kernel $K(\cdot)$ such that $K(\cdot) = K^\pm(\cdot) = K^\pm_\kappa(\cdot)$. To assess the effect of different memory structures, we compare three kernel types for $K(\cdot)$: exponential / Ornstein–Uhlenbeck (OU), Gaussian, and fractional. Each kernel encodes a distinct decay profile in how past trades affect current intensities

[4] Specifically, each value of $\{t_i, z_i\}_{i=1}^{M_{\text{int}}}$ is generated by first randomly and uniformly sampling $(t, x, q, \boldsymbol{\Lambda}^+_\lambda, \boldsymbol{\Lambda}^-_\lambda, \boldsymbol{\Lambda}^+_\kappa, \boldsymbol{\Lambda}^-_\kappa)$ from $[0,T) \times \mathbb{R} \times \{\underline{q}, \underline{q}+1, ..., \bar{q}-1, \bar{q}\} \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^N$, and then we set the values of $\lambda^\pm$ and $\kappa^\pm$ via the definitions $\lambda^\pm = \lambda_0^\pm(t) + a_\lambda^\pm \Lambda_\lambda^\pm$ and $\kappa^\pm = \kappa_0^\pm(t) + a_\kappa^\pm \, \Lambda_\kappa^\pm$. Similar approach is used for generating the values of $\{z_i^T\}_{i=1}^{M_{\text{term}}}$.

and liquidity, as listed in Table 1. Th other model parameters used in our numerical experiment are given as below:

$$\lambda_0^{\pm}(t) = 30.0, \qquad \kappa_0^{\pm} = 100, \qquad \underline{q} = -10, \qquad \bar{q} = 10,$$
$$\underline{\Lambda}_\kappa = -20, \qquad \bar{\Lambda}_\kappa = 20, \qquad \underline{\Lambda}_\lambda = -20, \qquad \bar{\Lambda}_\lambda = 20,$$
$$a_{\lambda^\pm} = 2.0, \qquad a_{\kappa^\pm} = 2.0, \qquad \alpha = 0.001, \qquad \beta = 0.01,$$
$$\epsilon^+ = 0.005, \qquad \epsilon^- = 0.005, \qquad T = 1.$$

Here, $\underline{\Lambda}$ and $\bar{\Lambda}$ specify the sampling ranges for the lifted states from which we draw data points to train the PINN.

Kernel	$K(t,s)$	**Parameters**
Exponential (OU)	$\exp\big(-\lvert t-s\rvert/a\big)$	$a = 0.1$
Gaussian	$\exp\big(-(t-s)^2/(2b^2)\big)$	$b = 0.316$
Fractional	$(1+\lvert t-s\rvert)^{-r}$	$r = 0.51$

Table 1: List of kernels to be considered in the numerical experiment.

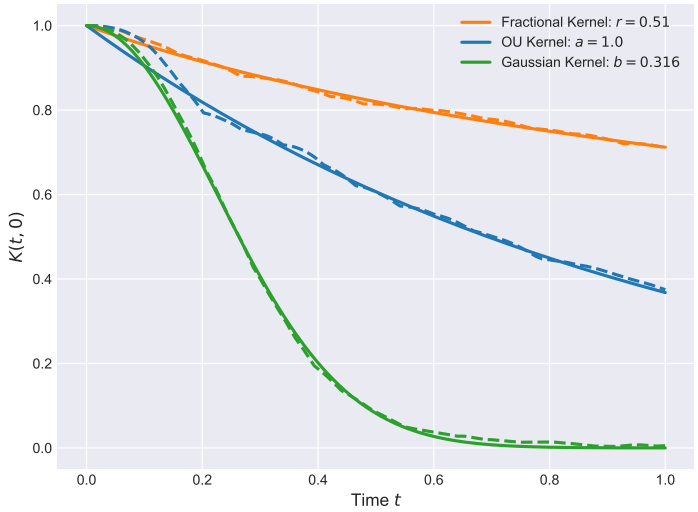


Figure 1: Plots of the kernels listed in Table 1. The solid lines correspond to the exact kernels $K(t,0)$, while the dashed lines represent the neural network approximations $\phi_{\hat{\boldsymbol{\theta}}}(t)^\top \phi_{\hat{\boldsymbol{\theta}}}(0)$.

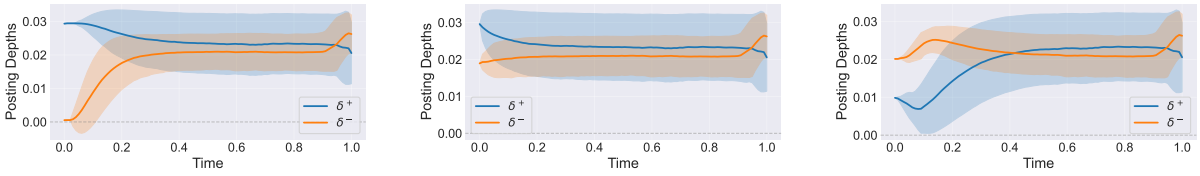
The decay-behaviour of each kernel is shown in Figure 1: the OU kernel exhibits rapid exponential decay, while the fractional kernel decays more slowly, retaining longer memory of past activity. The Gaussian kernel gives a bell-shaped decay which is somewhat a middle ground between the OU and fractional kernel. For each kernel, their feature map representation is learnt by a feedforward neural network with 4-dimensional outputs (i.e. $n = 4$), 1 hidden layer with ReLU activation and 64 neurons. The training is done against $M_{\text{kernel}} = 25000$ number of sample points. The learnt kernels are displayed in Figure 1 as well.

The PINN which represents the value function has 1 hidden layer consisting of 32 neurons with ReLU activation. The number of data points used for training are $M_{\text{int}} = M_{\text{term}} = 100000$, and the regularisation parameter is $\gamma = 3 \times 10^{-4}$. Trainings of all the neural networks in our experiment are done by gradient descent with Adam optimiser, gradient clipping and Xavier initialization.

4.4 The numerical results

We generate 1000 paths over the fixed horizon $T = 1$ and the market-maker posts bid and ask spreads dynamically according to the feedback control derived from the learnt value function. For each generated path indexed by j, each trajectory of the corresponding bid/ask spread is denoted by $(\delta_j^{\pm}(t))_{t\geq 0}$. In Figure 2 to 5, we plot the average values of $\delta_j^{\pm}(t)$ (and indicate the ranges of ± 1 standard deviation around the average values via the shaded region) across all the simulated paths $j \in \{1, .., 1000\}$ as a function of time t.

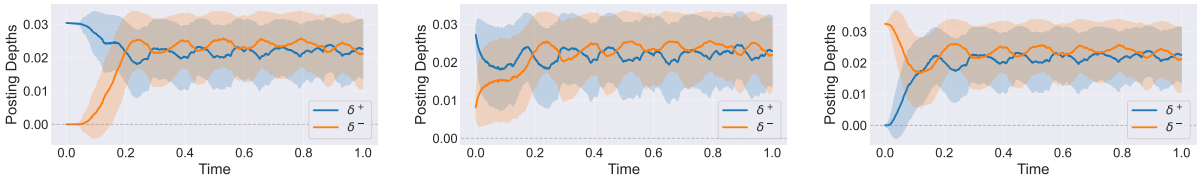
Figure 2 covers a degenerate setup without kernels which is equivalent to $a_{\lambda^{\pm}} = a_{\kappa^{\pm}} = 0$, i.e. there is no self-exciting behaviours with the market order flow and liquidity. The posted spreads are initially skewed depending on the starting inventory. In the case of negative (resp. positive) $Q(0)$, δ^+ is initially large (resp. small) and δ^- is initially small (resp. large) which are equivalent to high (resp. low) bid and ask prices. This is driven by the market-maker's incentive to keep the inventory level neutral. The spreads are stabilised quickly as the trading session goes on once the inventory balance has been restored.



(a) $Q(0) = -3$ (b) $Q(0) = 0$ (c) $Q(0) = 3$

Figure 2: Optimal posting depth paths for different initial inventories with no kernel (i.e. $a_{\lambda^{\pm}} = a_{\kappa^{\pm}} = 0$).

Next, we look at the case with self-exciting order flow and liquidity under OU kernels. We observe in Figure 3 that the posting depths are of similar magnitude to the no-kernel case, but the quotes now exhibit significant oscillations. When a market order arrives, the market-maker is aware of the jumps in the order arrival intensity as well as liquidity level. They therefore react by quickly adjusting their quotes. Meanwhile, they also anticipate that the influence from the past market orders will dissipate quickly under the OU kernel, and thus they will wind down their adjustment after a short duration. Consequently, the market-marker's quotes will oscillate over time as a result of their continual adjustment in presence of self-exciting order flow and liquidity under weak persistence.



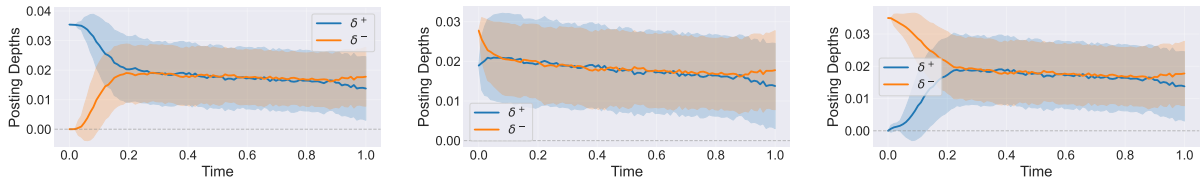
(a) $Q(0) = -3$ (b) $Q(0) = 0$ (c) $Q(0) = 3$

Figure 3: Optimal posting depths for different inventories with an OU kernel ($a = 0.1$).

More generally, the qualitative features of the quotes depend crucially on the kernels used. Under the fractional kernels, the shocks brought by the past market orders dissipate much slower and therefore the market-maker will adjust their quotes more gradually and overall the quotes are more stable over the trading session, as shown in Figure 4. For the Gaussian kernel, its persistence level is in-between that of the OU and fractional kernel. The quotes in Figure 5 therefore exhibit mild oscillation with relatively stable trajectories.

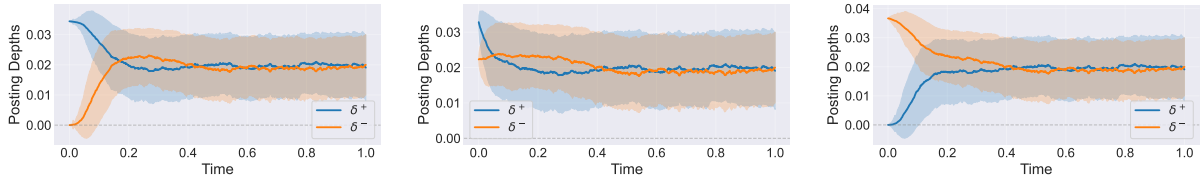
In Figure 6 and Figure 7, we focus again on the OU kernels but under several different parameter

a. A larger value of a results in a higher persistence level in that $K(t,s) = K(t-s,0)$ decays more slowly as the time lapse $t-s$ increases. See Figure 6. As the parameter a increases, we observe in Figure 7 that the quoted spreads become less oscillatory and tend to decrease slightly over time. There are two competing factors driving the direction of adjustment of the market-marker's spreads. On the one hand, the clustering of market orders suggest that that number of market orders will increase on average following an incoming market order which in turn improves the fill probability of the market-maker's order. On the other hand, the increase in the overall liquidity level triggered by an incoming market order will reduce the conditional fill probability of the market-maker's limit order because of the improved liquidity provision elsewhere. In this particular numerical example, the effect from the rise in liquidity level dominates under high persistence parameter. The market-maker therefore has to provide shallower quotes to remain competitive.



(a) $Q(0) = -3$ (b) $Q(0) = 0$ (c) $Q(0) = 3$

Figure 4: Optimal posting depths for different inventories with a fractional kernel ($r = 0.51$).



(a) $Q(0) = -3$ (b) $Q(0) = 0$ (c) $Q(0) = 3$

Figure 5: Optimal posting depths for different inventories with a Gaussian kernel ($b = 0.316$).

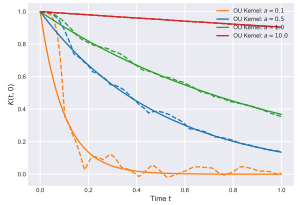
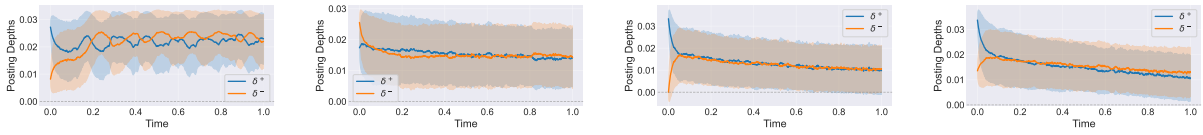


Figure 6: OU kernel shapes for varying a.

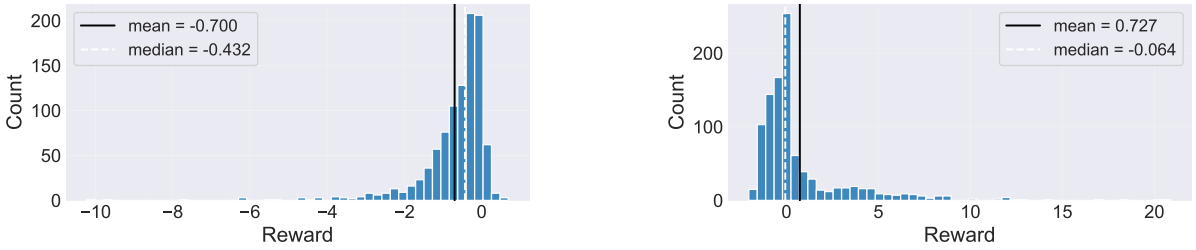


(a) $a = 0.1$ (b) $a = 0.5$ (c) $a = 1.0$ (d) $a = 10.0$

Figure 7: Posting depths for $Q(0) = 0$ under OU kernels with varying a.

Finally, we illustrate the importance of taking the self-exciting behaviours of the order flow and

liquidity level into the account. Under the same baseline parameters, we numerically estimate two sets of market-making strategies, one learnt using the OU feature map with $a = 1.0$, and one learnt as if there is no kernel (i.e. $a_{\lambda^\pm} = a_{\kappa^\pm} = 0$). We then simulate the trading environment where the prices and order flows are driven by the OU kernels with $a = 1.0$ and evaluate the performance attained by each set of the strategies. In simple terms, the first set of strategy represents the one pursued by a sophisticated market-maker who correctly identifies the price and order flow dynamics, whereas the second strategy is the one pursed by a naive/myopic market-maker who fails to anticipate the self-exciting behaviours of the order flow and liquidity.



(a) Naive market-maker.

(b) Sophisticated market-maker.

Figure 8: The empirical distributions of the realised performance $X_T + Q_T(S_T - \alpha Q_T) - \beta \int_0^T (Q_u)^2 du$ an OU environment with $a = 1.0$. Panel (a) is produced by a "naive strategy" where the market-maker determines the optimal spread (δ^+, δ^-) as if there is no kernel, i.e. $K(\cdot) \equiv 1$. Panel (b) is produced by the optimal strategy estimated via the procedure in Section 4.2.

The empirical distributions of the performance (the terminal profit-and-loss penalised by inventory holdings) attained by the two market-makers are shown in Figure 8. The sophisticated market-maker achieves a higher mean reward compared to the naive one. Moreover, we also note that the performance of the sophisticated market-maker is positively skewed while that of the naive one is negatively skewed, indicating that the appropriate incorporation of the kernels not only improves the average performance but also helps controlling the downside risk of the market-maker's financial position.

5 Conclusion and further work

We develop a market-making framework that links persistence in order flow and market liquidity to optimal quoting strategies, in which a full-fledged Hawkes process is considered to model the order arrival intensity as well as the stochastic liquidity level, where the latter influences the conditional filled probability of a market-marker's limit order. Our theoretical contribution is to propose a Markovian approximation approach via estimating the memory kernels with the truncated Mercer expansions, and we show that the approximated value function theoretically converges to the actual value function. On the methodological front, we demonstrate how the finite-dimensional "lifted version" of the stochastic control problem can be practically solved using a neural network approximation of the kernels in conjunction with a PINN to tackle the resulting PDE. Our numerical results show that kernel persistence plays a central role in shaping optimal strategies, where the quotation behaviours can be drastically different under different parameterisation of the kernel. We also show that proper incorporation of flow and liquidity persistence is financially important to improve trading performance.

In terms of avenue of future research, one could explore stochastic baseline intensities, cross-excitation between buy and sell flows, or reinforcement learning methods that build directly on the Mercer's representation. The action space could also be expanded by allowing the trader to place both limit order and market order such that the problem would become a hybrid of market-making and optimal execution

problem. Such extensions would further bridge the gap between high-frequency market microstructure and practical algorithmic trading design.

Disclosure of interest

We have no competing interest to declare.

Declaration of funding

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Appendix A Convergence of the approximated value function

This entire appendix is dedicated to the proof of Theorem 2.4. The ideas behind the proof are based on the constructions and analysis of a sequence of Hawkes processes living on a common probability space. In what follows, we will reintroduce our modelling setup and define several new notations.

We work with a fixed terminal horizon $T > 0$ and a fixed filtered probability space $(\Omega, \{\mathcal{F}_t\}_{t\leq T}, \mathcal{F}, \mathbb{P})$ that supports a Brownian motion $W(\omega; t) : \Omega \times [0, T] \to \mathbb{R}$, and two Poisson random measures on $[0, T] \times \mathbb{R}_+$ denoted by $\mathcal{N}^+(\omega; dt, du) : \Omega \times \mathcal{B}([0, T] \times \mathbb{R}_+) \to \mathbb{Z}_+$ and $\mathcal{N}^-(\omega; dt, du) : \Omega \times \mathcal{B}([0, T] \times \mathbb{R}_+) \to \mathbb{Z}_+$ where each of them has a Lebesgue intensity measure. The natural filtrations generated by W and $\mathcal{N}^\pm$ are $\mathbb{F}^W = \{\mathcal{F}_t^W\}_{t\leq T}$ and $\mathbb{F}^{\mathcal{N}^\pm} = \{\mathcal{F}_t^{\mathcal{N}^\pm}\}_{t\leq T}$ respectively, with $\mathcal{F}_t^W := \sigma(W_s : s \leq t)$ and $\mathcal{F}_t^{\mathcal{N}^\pm} := \sigma(\mathcal{N}^\pm(A) : A \in \mathcal{B}([0, t] \times \mathbb{R}_+))$. We assume W, $\mathcal{N}^+$ and $\mathcal{N}^-$ are jointly independent.

Without loss of generality, we assume the model parameters of the market-making problem are such that $a_{\lambda^\pm} = a_{\kappa^\pm} = 1$. The general case can be analysed easily by scaling the memory kernels appropriately.

For some fixed $\rho \in (0, 1)$, write $\mathcal{D} := \{(t, s) \in [0, T]^2 : s \leq t\}$ and define

$$\mathcal{H}_\rho := \left\{ f \in \mathcal{C}(\mathcal{D}) : \max\left(\sup_{t\in[0,T]} \int_0^t |f(t, s)|ds, \sup_{s\in[0,T]} \int_s^T |f(t, s)|dt \right) < \rho \right\}. \tag{A.1}$$

For some fixed $\Lambda \in (0, \infty)$, let $\mathcal{L}_\Lambda$ be the set of all measurable functions $f : [0, T] \to \mathbb{R}_{++}$ with $\sup_{t\leq T} f(t) \leq \Lambda$. Introduce four deterministic functions $(\lambda_0^+, \lambda_0^-, \kappa_0^+, \kappa_0^-) \in \mathcal{L}_\Lambda^4$ which represent the baseline intensity of the market order flow processes and the baseline liquidity processes.

A.1 Construction of a parameterised family of the state processes

On our reference filtered probability space, we now construct a family of the state processes arising in our market-making problem parameterised by $(K^+, K^-, K_\kappa^+, K_\kappa^-) \in \mathcal{H}_\rho^4$. The unique solutions (M^+, M^-) to

$$M^\pm(t) = \int_0^t \int_0^\infty \mathbb{1}_{\{u\leq\lambda^\pm(s-)\}} \mathcal{N}^\pm(ds, du) \tag{A.2}$$

with

$$\lambda^\pm(t) = \lambda_0^\pm(t) + \int_0^t K^\pm(t, s) dM^\pm(s) \tag{A.3}$$

define two Hawkes processes with baseline deterministic intensity functions $\lambda_0^{\pm}$ and memory kernels $K^{\pm}$.[5] By the compensator identity and the definitions of $\mathcal{H}_\rho$ and $\mathcal{L}_\Lambda$, we have

$$\mathbb{E}[\lambda^{\pm}(t)] = \lambda_0^{\pm}(t) + \int_0^t K^{\pm}(t,s)\mathbb{E}[\lambda^{\pm}(s)]ds \leq \Lambda + \rho \sup_{s\leq t} \mathbb{E}[\lambda^{\pm}(s)]$$

for all $t \in [0,T]$, from which we can deduce

$$\mathbb{E}[\lambda^{\pm}(t)] \leq \sup_{s\leq t} \mathbb{E}[\lambda^{\pm}(s)] \leq \frac{\Lambda}{1-\rho}, \qquad t \in [0,T]. \tag{A.4}$$

By following similar arguments in Leblanc (2024), one can show that the constructed Hawkes processes have finite exponential moment. Consequently, for any $n \in \mathbb{N}$ and $t \in [0,T]$ there exists some deterministic $A_{\Lambda,\rho,n}(t) \in (0,\infty)$ depending on (Λ, ρ, n, t) only such that

$$\mathbb{E}[(M^{\pm}(t))^n] \leq A_{\Lambda,\rho,n}(t). \tag{A.5}$$

Let $\tau_i^{\pm}$ be the ith jump time of $M^{\pm}$, i.e.

$$\tau_i^{\pm} := \inf\{t \geq 0 : M^{\pm}(t) = i\}, \qquad i \in \mathbb{N}.$$

By convention, we set $\tau_0^{\pm} := 0$. The underlying stock price process S is given by

$$S(t) = S_0 + \sigma W(t) + \epsilon^+ M^+(t) - \epsilon^- M^-(t).$$

Next, the pair of stochastic liquidity proxy processes (κ^+, κ^-) follows the dynamics

$$\kappa^{\pm}(t) = \kappa_0^{\pm}(t) + \int_0^t K_\kappa^{\pm}(t,s) dM^{\pm}(s),$$

while the conditional filled probability processes (p^+, p^-) are

$$p^{\pm}(t) = \exp\left(-\kappa^{\pm}(t)\delta^{\pm}(t)\right)$$

with $\delta = (\delta^+, \delta^-) \in \mathcal{A}$, where the admissible set $\mathcal{A}$ is defined in (2.5). The net inventory process Q, the cumulative inventory purchased process N^- and the cumulative inventory sold process N^+ are defined via

$$Q(t) = q_0 + N^-(t) - N^+(t),$$

with

$$N^+(t) = \int_0^t \int_0^\infty \mathbb{1}_{\{u \leq \lambda^+(s-)p^+(s-), Q(s-) > \underline{q}\}} \mathcal{N}^+(\omega; ds, du),$$
$$N^-(t) = \int_0^t \int_0^\infty \mathbb{1}_{\{u \leq \lambda^-(s-)p^-(s-), Q(s-) < \bar{q}\}} \mathcal{N}^-(\omega; ds, du),$$

where the constants $\underline{q} < 0 < \bar{q}$ represent the long/short position limits on the net inventory. Recall that we allow $\underline{q} = -\infty$ or $\bar{q} = +\infty$ which then correspond to a special case of no upper and/or lower position limit. Finally, the cash process is

$$X(t) = x_0 + \int_0^t (S(u-) + \delta^+(u)) dN^+(u) - \int_0^t (S(u-) - \delta^-(u)) dN^-(u)$$

for some given $x_0 \in \mathbb{R}$.

[5] The construction of a Hawkes process (and its extensions) via such techniques of Poisson random measure embedding is well-known in the literature. See for example Brémaud and Massoulié (1996) and Hillairet et al. (2023).

All the quantities/processes constructed above can be compactly summarised by a tuple with parameters $(K^+, K^-, K^+_\kappa, K^-_\kappa) \in \mathcal{H}^4_\rho$ as

$$Y = Y(K^+, K^-, K^+_\kappa, K^-_\kappa) = ((M^\pm(t))_{t\geq 0}, (S(t))_{t\geq 0}, (\lambda^\pm(t))_{t\geq 0}, (\kappa^\pm(t))_{t\geq 0}, (p^\pm(t))_{t\geq 0}, \\ \{\tau_i^\pm\}_{i\in\mathbb{N}}, (N^\pm(t))_{t\geq 0}, (Q(t))_{t\geq 0}, (X(t))_{t\geq 0}).$$

In the sequel, we will consider a family of Y on the same probability space as $(K^+, K^-, K^+_\kappa, K^-_\kappa)$ varies. Importantly, different copies of $Y = Y(\omega)$ are all constructed pathwise against the same $W(\omega, \cdot)$ and $\mathcal{N}^\pm(\omega; \cdot, \cdot)$.

A.2 Some estimates and the uniform convergence of the approximating processes in expectation

Fix a sequence of kernels $\{(K^+_n, K^-_n, K^+_{\kappa,n}, K^-_{\kappa,n})\}_{n\in\mathbb{N}}$ where $(K^+_n, K^-_n, K^+_{\kappa,n}, K^-_{\kappa,n}) \in \mathcal{H}^4_\rho$ for each n. We also fix a reference set of kernels $(K^+, K^-, K^+_\kappa, K^-_\kappa) \in \mathcal{H}_\rho$. On the same reference filtered probability space, construct

$$\begin{aligned} Y &= Y(K^+, K^-, K^+_\kappa, K^-_\kappa) \\ &= ((M^\pm(t))_{t\geq 0}, (S(t))_{t\geq 0}, (\lambda^\pm(t))_{t\geq 0}, (\kappa^\pm(t))_{t\geq 0}, (p^\pm(t))_{t\geq 0}, \{\tau_i^\pm\}_{i\in\mathbb{N}}, (N^\pm(t))_{t\geq 0}, (Q(t))_{t\geq 0}, (X(t))_{t\geq 0}) \end{aligned}$$

and for each n

$$\begin{aligned} Y_n &= Y(K^+_n, K^-_n, K^+_{\kappa,n}, K^-_{\kappa,n}) \\ &= ((M^\pm_n(t))_{t\geq 0}, (S_n(t))_{t\geq 0}, (\lambda^\pm_n(t))_{t\geq 0}, (\kappa^\pm_n(t))_{t\geq 0}, (p^\pm_n(t))_{t\geq 0}, \{\tau^\pm_{i,n}\}_{i\in\mathbb{N}}, (N^\pm_n(t))_{t\geq 0}, (Q_n(t))_{t\geq 0}, (X_n(t))_{t\geq 0}) \end{aligned}$$

according to Section A.1. For $t \in [0, T]$ and $n \in \mathbb{N}$, define

$$\begin{aligned} \epsilon^\pm_n(t) &:= \int_0^t |K^\pm(t,u) - K^\pm_n(t,u)|du, \\ E^\pm_n(t) &:= \sup_{u\in[0,t]} \int_u^t |K^\pm(s,u) - K^\pm_n(s,u)|ds, \\ E^\pm_{\kappa,n} &:= \sup_{(t,s)\in\mathcal{D}} |K^\pm_\kappa(t,s) - K^\pm_{\kappa,n}(t,s)|. \end{aligned}$$

Proposition A.1. *The following estimates hold:*

$$\mathbb{E}\left[\sup_{0\le u\le t}|M^{\pm}(u)-M_n^{\pm}(u)|\right]\le\frac{\Lambda}{(1-\rho)^2}\min\left(E_n^{\pm}(t),\int_0^t\epsilon_n^{\pm}(s)ds\right), \tag{A.6}$$

$$\mathbb{E}\left[\left(\sup_{0\le u\le t}|M^{\pm}(u)-M_n^{\pm}(u)|\right)^2\right]\le\frac{1}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n^{\pm}(t)+A_{\Lambda,\rho,2}(t)(E_n^{\pm}(t))^2+3\rho E_n^{\pm}(t)A_{\Lambda,\rho,2}(t)\right], \tag{A.7}$$

$$\mathbb{E}\left[\int_0^t|\lambda^{\pm}(u)-\lambda_n^{\pm}(u)|du\right]\le\frac{\Lambda}{(1-\rho)^2}E_n^{\pm}(t), \tag{A.8}$$

$$\begin{aligned}\mathbb{E}\left[\left(\int_0^t|\lambda^{\pm}(u)-\lambda_n^{\pm}(u)|du\right)^2\right]&\le A_{\Lambda,\rho,2}(t)(E_n^{\pm}(t))^2+3\rho E_n^{\pm}(t)A_{\Lambda,\rho,2}(t)\\&\quad+\frac{\rho^2}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n^{\pm}(t)+A_{\Lambda,\rho,2}(t)(E_n^{\pm}(t))^2+3\rho E_n^{\pm}(t)A_{\Lambda,\rho,2}(t)\right],\end{aligned} \tag{A.9}$$

$$\mathbb{E}\left[\sup_{0\le u\le t}|\kappa^{\pm}(u)-\kappa_n^{\pm}(u)|\right]\le\frac{\Lambda t}{1-\rho}E_{\kappa,n}^{\pm}+\frac{\Lambda\rho}{(1-\rho)^2}E_n^{\pm}(t), \tag{A.10}$$

$$\mathbb{E}\left[\sup_{0\le u\le t}|p^{\pm}(u)-p_n^{\pm}(u)|\right]\le C\left(\frac{\Lambda t}{1-\rho}E_{\kappa,n}^{\pm}+\frac{\Lambda\rho}{(1-\rho)^2}E_n^{\pm}(t)\right), \tag{A.11}$$

$$\begin{aligned}\mathbb{E}\left[\left(\sup_{0\le u\le t}|S(u)-S_n(u)|\right)^2\right]&\le\frac{2(\epsilon_+)^2}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n^{+}(t)+A_{\Lambda,\rho,2}(t)(E_n^{+}(t))^2+3\rho E_n^{+}(t)A_{\Lambda,\rho,2}(t)\right]\\&\quad+\frac{2(\epsilon^-)^2}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n^{-}(t)+A_{\Lambda,\rho,2}(t)(E_n^{-}(t))^2+3\rho E_n^{-}(t)A_{\Lambda,\rho,2}(t)\right],\end{aligned} \tag{A.12}$$

where $A_{\Lambda,\rho,2}(t)>0$ is the moment estimate of $\mathbb{E}[(M^{\pm}(t))^2]$ as per (A.5) *and $C>0$ is the upper bound of the admissible spreads arising in the definition of $\mathcal{A}$ in* (2.5).

Proof. Suppress the superscripts $\pm$ everywhere for brevity. By (A.2),

$$\begin{aligned}|M(t)-M_n(t)|&=\left|\int_0^t\int_0^{\infty}(\mathbb{1}_{\{u\le\lambda(s-)\}}-\mathbb{1}_{\{u\le\lambda_n(s-)\}})\mathcal{N}(ds,du)\right|\\&\le\int_0^t\int_0^{\infty}|\mathbb{1}_{\{u\le\lambda(s-)\}}-\mathbb{1}_{\{u\le\lambda_n(s-)\}}|\mathcal{N}(ds,du).\end{aligned} \tag{A.13}$$

Since the right-hand-side of (A.13) is increasing in t, we have

$$\sup_{u\le t}|M(u)-M_n(u)|\le\int_0^t\int_0^{\infty}|\mathbb{1}_{\{u\le\lambda(s-)\}}-\mathbb{1}_{\{u\le\lambda_n(s-)\}}|\mathcal{N}(ds,du)$$

for all $t\le T$. Recall that $\mathcal{N}$ is a Poisson measure with Lebesgue intensity measure. The right-hand-side of the above is, conditional on the realisation of $(\lambda(s),\lambda_n(s))_{s\le t}$, a Poisson random variable with parameter

$$\int_0^t\int_0^{\infty}|\mathbb{1}_{\{u\le\lambda(s-)\}}-\mathbb{1}_{\{u\le\lambda_n(s-)\}}|duds=\int_0^t|\lambda(s)-\lambda_n(s)|ds.$$

Hence we have

$$\mathbb{E}\left[\sup_{u\le t}|M(u)-M_n(u)|\right]\le\mathbb{E}\left[\int_0^t|\lambda(s)-\lambda_n(s)|ds\right]. \tag{A.14}$$

and

$$\mathbb{E}\left[\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2\right]\le\mathbb{E}\left[\int_0^t|\lambda(s)-\lambda_n(s)|ds\right]+\mathbb{E}\left[\left(\int_0^t|\lambda(s)-\lambda_n(s)|ds\right)^2\right]. \tag{A.15}$$

Now, by (A.3),

$$|\lambda(t) - \lambda_n(t)| = \left| \int_0^t K(t,s) dM(s) - \int_0^t K_n(t,s) dM_n(s) \right|$$
$$\leq \int_0^t |K(t,s) - K_n(t,s)| dM(s) + \int_0^t K_n(t,s) d|M - M_n|(s),$$

where the last term is understood as the integration of $s \mapsto K_n(t,s)$ against the pure jump process $s \mapsto |M(s) - M_n(s)|$. Further integration on the both sides of the above leads to

$$\int_0^t |\lambda(s) - \lambda_n(s)| ds \leq \int_0^t \int_0^s |K(s,u) - K_n(s,u)| dM(u) ds + \int_0^t \int_0^s K_n(s,u) d|M - M_n|(u) ds \quad \text{(A.16)}$$
$$= \int_0^t \left(\int_u^t |K(s,u) - K_n(s,u)| ds \right) dM(u) + \int_0^t \left(\int_u^t K_n(s,u) ds \right) d|M - M_n|(u)$$
$$\leq \int_0^t \left(\sup_{u \leq t} \int_u^t |K(s,u) - K_n(s,u)| ds \right) dM(u) + \rho \int_0^t d|M - M_n|(u)$$
$$= E_n(t) M(t) + \rho |M(t) - M_n(t)| \quad \text{(A.17)}$$
$$\leq E_n(t) M(t) + \rho \sup_{u \leq t} |M(u) - M_n(u)|,$$

where we have used Fubini's theorem, the definition of $E_n(t)$ and the fact that $K_n \in \mathcal{H}_\rho$. Taking expectation, we have

$$\int_0^t \mathbb{E}|\lambda(s) - \lambda_n(s)| ds \leq E_n(t) \mathbb{E}[M(t)] + \rho \mathbb{E}\left[\sup_{u \leq t} |M(t) - M_n(t)| \right]$$
$$\leq \frac{\Lambda}{1-\rho} E_n(t) + \rho \mathbb{E}\left[\sup_{u \leq t} |M(t) - M_n(t)| \right] \quad \text{(A.18)}$$

with the help of (A.4). Combining (A.14) and (A.18) leads to

$$\mathbb{E}\left[\sup_{u \leq t} |M(t) - M_n(t)| \right] \leq \frac{\Lambda}{1-\rho} E_n(t) + \rho \mathbb{E}\left[\sup_{u \leq t} |M(t) - M_n(t)| \right]$$

and hence

$$\mathbb{E}\left[\sup_{u \leq t} |M(t) - M_n(t)| \right] \leq \frac{\Lambda}{(1-\rho)^2} E_n(t).$$

Alternatively, if we take expectation on the both sides of (A.16) instead, we will obtain

$$\mathbb{E}\left[\int_0^t |\lambda(s) - \lambda_n(s)| ds \right] \leq \mathbb{E}\left[\int_0^t \int_0^s |K(s,u) - K_n(s,u)| dM(u) ds \right] + \rho \mathbb{E}\left[\sup_{u \leq t} |M(u) - M_n(u)| \right]$$
$$= \int_0^t \mathbb{E}\left[\int_0^s |K(s,u) - K_n(s,u)| dM(u) \right] ds + \rho \mathbb{E}\left[\sup_{u \leq t} |M(u) - M_n(u)| \right]$$
$$= \int_0^t \mathbb{E}\left[\int_0^s |K(s,u) - K_n(s,u)| \lambda(u) du \right] ds + \rho \mathbb{E}\left[\sup_{u \leq t} |M(u) - M_n(u)| \right]$$
$$\leq \int_0^t \int_0^s \left(|K(s,u) - K_n(s,u)| \mathbb{E}[\lambda(u)] du \right) ds + \rho \mathbb{E}\left[\sup_{u \leq t} |M(u) - M_n(u)| \right]$$
$$\leq \frac{\Lambda}{1-\rho} \int_0^t \epsilon_n(s) ds + \rho \mathbb{E}\left[\sup_{u \leq t} |M(u) - M_n(u)| \right],$$

where we have used the compensator identity, the definition of $\epsilon_n(\cdot) = \epsilon_n^{\pm}(\cdot)$ and (A.4). The above inequality together with (A.14) leads to

$$\mathbb{E}\left[\sup_{u \leq t} |M(t) - M_n(t)| \right] \leq \frac{\Lambda}{(1-\rho)^2} \int_0^t \epsilon_n(s) ds.$$

This verifies the estimates in (A.6).

On squaring both sides of (A.17), we arrive at

$$\begin{aligned}\left(\int_0^t |\lambda(s)-\lambda_n(s)|ds\right)^2 &\le E_n^2(t)M^2(t)+2\rho E_n(t)M(t)|M(t)-M_n(t)|+\rho^2|M(t)-M_n(t)|^2\\ &\le E_n^2(t)M^2(t)+2\rho E_n(t)M(t)|M(t)-M_n(t)|+\rho^2\sup_{u\le t}|M(u)-M_n(u)|^2\\ &\le E_n^2(t)M^2(t)+\rho E_n(t)[M^2(t)+|M(t)-M_n(t)|^2]+\rho^2\sup_{u\le t}|M(u)-M_n(u)|^2\\ &\le E_n^2(t)M^2(t)+\rho E_n(t)[2M^2(t)+M_n^2(t)]+\rho^2\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2,\end{aligned}$$

where we note that $\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2=\sup_{u\le t}\left(|M(u)-M_n(u)|^2\right)$ because $|M(u)-M_n(u)|\in\mathbb{Z}_+$ for all u. Hence

$$\begin{aligned}\mathbb{E}\left[\left(\int_0^t|\lambda(s)-\lambda_n(s)|ds\right)^2\right] &\le E_n^2(t)\mathbb{E}[M^2(t)]+\rho E_n(t)\left(2\mathbb{E}[M^2(t)]+\mathbb{E}[M_n^2(t)]\right)+\rho^2\mathbb{E}\left[\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2\right]\\ &\le A_{\Lambda,\rho,2}(t)E_n^2(t)+3\rho E_n(t)A_{\Lambda,\rho,2}(t)+\rho^2\mathbb{E}\left[\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2\right].\end{aligned}\tag{A.19}$$

On combining the above estimate with (A.15), (A.18) and (A.6), we have

$$\begin{aligned}\mathbb{E}\left[\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2\right] &\le \frac{\Lambda}{1-\rho}E_n(t)+\rho\frac{\Lambda}{(1-\rho)^2}E_n(t)\\ &\quad+A_{\Lambda,\rho,2}(t)E_n^2(t)+3\rho E_n(t)A_{\Lambda,\rho,2}(t)+\rho^2\mathbb{E}\left[\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2\right]\end{aligned}$$

and therefore

$$\mathbb{E}\left[\left(\sup_{u\le t}|M(u)-M_n(u)|\right)^2\right]\le\frac{1}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n(t)+A_{\Lambda,\rho,2}(t)E_n^2(t)+3\rho E_n(t)A_{\Lambda,\rho,2}(t)\right].$$

Now, (A.8) is simply a consequence of (A.18) and part (1) of this lemma, while (A.9) is simply a consequence of (A.19) and part (2) of this lemma.

Following similar analysis that we have considered for $|\lambda(t)-\lambda_n(t)|$, we have

$$\begin{aligned}|\kappa^\pm(t)-\kappa_n^\pm(t)| &= \left|\int_0^t K_\kappa^\pm(t,s)dM^\pm(s)-\int_0^t K_{\kappa,n}^\pm(t,s)dM_n^\pm(s)\right|\\ &\le \int_0^t|K_\kappa^\pm(t,s)-K_{\kappa,n}^\pm(t,s)|dM^\pm(s)+\int_0^t K_{\kappa,n}(t,s)d|M-M_n|(s)\\ &\le \int_0^t|K_\kappa^\pm(t,s)-K_{\kappa,n}^\pm(t,s)|dM^\pm(s)+\rho\sup_{u\le t}|M(u)-M_n(u)|\end{aligned}$$

and hence

$$\begin{aligned}\mathbb{E}\left[\sup_{u\le t}|\kappa^\pm(u)-\kappa_n^\pm(u)|\right] &\le \mathbb{E}\left[\int_0^t|K_\kappa^\pm(t,s)-K_{\kappa,n}^\pm(t,s)|\lambda^\pm(s)ds\right]+\rho\mathbb{E}\left[\sup_{u\le t}|M(u)-M_n(u)|\right]\\ &\le E_{\kappa,n}^\pm\int_0^t\mathbb{E}[\lambda^\pm(s)]ds+\rho\mathbb{E}\left[\sup_{u\le t}|M(u)-M_n(u)|\right]\\ &\le \frac{\Lambda t}{1-\rho}E_{\kappa,n}^\pm+\frac{\Lambda\rho}{(1-\rho)^2}E_n^\pm.\end{aligned}$$

Thus (A.10) holds.

By the fundamental inequality $|e^{-x} - e^{-y}| \le |x - y|$ for $x, y \ge 0$, we have

$$|p^{\pm}(t) - p_n^{\pm}(t)| = |\exp\left(-\kappa^{\pm}(t)\delta^{\pm}(t)\right) - \exp\left(-\kappa_n^{\pm}(t)\delta^{\pm}(t)\right)| \le C|\kappa^{\pm}(t) - \kappa_n^{\pm}(t)|$$

on recalling the assumption that $\delta^{\pm}$ are non-negative and bounded by a constant C. Then

$$\begin{aligned}\mathbb{E}\left[\sup_{0\le u\le t}|p^{\pm}(u) - p^{\pm}(u)|\right] &\le C\mathbb{E}\left[\sup_{0\le u\le t}|\kappa^{\pm}(u) - \kappa_n^{\pm}(u)|\right] \\ &\le C\left(\frac{\Lambda t}{1-\rho}E_{\kappa,n}^{\pm} + \frac{\Lambda\rho}{(1-\rho)^2}E_n^{\pm}\right),\end{aligned}$$

and this verifies (A.11).

Finally,

$$|S(t) - S_n(t)| = |\epsilon^+(M^+(t) - M_n^+(t)) - \epsilon^-(M^-(t) - M_n^-(t))|$$

such that

$$\begin{aligned}\mathbb{E}\left[\left(\sup_{0\le u\le t}|S(u) - S_n(u)|\right)^2\right] &\le 2(\epsilon^+)^2\mathbb{E}\left[\left(\sup_{u\le t}|M^+(u) - M_n^+(u)|\right)^2\right] + 2(\epsilon^-)^2\mathbb{E}\left[\left(\sup_{u\le t}|M^-(u) - M_n^-(u)|\right)^2\right] \\ &\le \frac{2(\epsilon_+)^2}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n^+(t) + A_{\Lambda,\rho,2}(t)(E_n^+(t))^2 + 3\rho E_n^+(t)A_{\Lambda,\rho,2}(t)\right] \\ &\quad + \frac{2(\epsilon^-)^2}{1-\rho^2}\left[\frac{\Lambda}{(1-\rho)^2}E_n^-(t) + A_{\Lambda,\rho,2}(t)(E_n^-(t))^2 + 3\rho E_n^-(t)A_{\Lambda,\rho,2}(t)\right].\end{aligned}$$

Hence (A.12) holds. □

Corollary A.2. *If it holds that*

$$\lim_{n\to\infty} E_n^{\pm}(T) = \lim_{n\to\infty} E_{\kappa,n}^{\pm} = 0, \tag{A.20}$$

then

1. $\lim_{n\to\infty}\mathbb{E}\left[\left(\sup_{0\le u\le T}|M^{\pm}(u) - M_n^{\pm}(u)|\right)^2\right] = 0$;
2. $\lim_{n\to\infty}\mathbb{E}\left[\left(\sup_{0\le u\le T}|S^{\pm}(u) - S_n^{\pm}(u)|\right)^2\right] = 0$;
3. $\lim_{n\to\infty}\mathbb{E}\left[\left(\int_0^T|\lambda^{\pm}(u) - \lambda_n^{\pm}(u)|du\right)^2\right] = 0$;
4. $\lim_{n\to\infty}\mathbb{E}\left[\sup_{0\le u\le T}|p(u)^{\pm} - p_n^{\pm}(u)|\right] = 0$.

Proof. These are the direct consequences of the estimates in Proposition A.1. □

Proposition A.3. *Suppose Condition A.20 holds, and assume further that $\underline{q} = -\infty$ and $\bar{q} = +\infty$, i.e. there is no inventory constraint. Then*

$$\lim_{n\to 0}\mathbb{E}\left[\left(\sup_{0\le u\le T}|N^{\pm}(u) - N_n^{\pm}(u)|\right)^2\right] = \lim_{n\to 0}\mathbb{E}\left[\left(\sup_{0\le u\le T}|Q(u) - Q_n(u)|\right)^2\right] = 0.$$

Proof. Suppress the superscript $\pm$ for brevity. Without the inventory constraint, we have

$$\begin{aligned}|N(t) - N_n(t)| &= \left|\int_0^t\int_0^{\infty}(\mathbb{1}_{\{u\le\lambda(s-)p(s-)\}} - \mathbb{1}_{\{u\le\lambda_n(s-)p_n(s-)\}})\mathcal{N}(ds,du)\right| \\ &\le \int_0^t\int_0^{\infty}|\mathbb{1}_{\{u\le\lambda(s-)p(s-)\}} - \mathbb{1}_{\{u\le\lambda_n(s-)p_n(s-)\}}|\mathcal{N}(ds,du)\end{aligned}$$

and hence

$$\sup_{u\leq T}|N(u)-N_n(u)|\leq \int_0^T\int_0^\infty |\mathbb{1}_{\{u\leq \lambda(s-)p(s-)\}}-\mathbb{1}_{\{u\leq \lambda_n(s-)p_n(s-)\}}|\mathcal{N}(ds,du).$$

Conditional on the realisation of $(\lambda(s),\lambda_n(s))_{s\leq t}$, the right-hand-side of the above is a Poisson random variable with parameter

$$H_n:=\int_0^T\int_0^\infty |\mathbb{1}_{\{u\leq \lambda(s-)p(s-)\}}-\mathbb{1}_{\{u\leq \lambda_n(s-)p_n(s-)\}}|duds=\int_0^T|\lambda(s-)p(s-)-\lambda_n(s-)p_n(s-)|ds.$$

We therefore have

$$\mathbb{E}\left[\left(\sup_{u\leq T}|N(u)-N_n(u)|\right)^2\right]\leq \mathbb{E}[H_n]+\mathbb{E}[H_n^2]. \tag{A.21}$$

To show that $\mathbb{E}\left[\left(\sup_{u\leq T}|N(u)-N_n(u)|\right)^2\right]\to 0$, it is therefore sufficient to show that $\mathbb{E}[H_n^2]\to 0$

Now observe that

$$\begin{aligned}
H_n&=\int_0^T|\lambda(s-)p(s-)-\lambda_n(s-)p_n(s-)|ds\\
&\leq \int_0^T\lambda(s)|p(s)-p_n(s)|ds+\int_0^T p_n(s)|\lambda(s)-\lambda_n(s)|ds\\
&\leq \sup_{s\leq t}|p(s)-p_n(s)|\int_0^T\lambda(s)ds+\int_0^T|\lambda(s)-\lambda_n(s)|ds.
\end{aligned} \tag{A.22}$$

On taking square on the both sides, we have

$$\begin{aligned}
H_n^2\leq &\left(\sup_{s\leq T}|p(s)-p_n(s)|\right)^2\left(\int_0^T\lambda(s)ds\right)^2+2\left(\int_0^T\lambda(s)ds\right)\left(\int_0^T|\lambda(s)-\lambda_n(s)|ds\right)\\
&+\left(\int_0^T|\lambda(s)-\lambda_n(s)|ds\right)^2.
\end{aligned}$$

Taking expectation and using the Cauchy-Schwarz inequality, we deduce

$$\begin{aligned}
\mathbb{E}[H_n^2]\leq &\left\{\mathbb{E}\left[\left(\sup_{s\leq T}|p(s)-p_n(s)|\right)^4\right]\mathbb{E}\left[\left(\int_0^T\lambda(s)ds\right)^4\right]\right\}^{\frac{1}{2}}\\
&+2\left\{\mathbb{E}\left[\left(\int_0^T\lambda(s)ds\right)^2\right]\mathbb{E}\left[\left(\int_0^T|\lambda(s)-\lambda_n(s)|ds\right)^2\right]\right\}^{\frac{1}{2}}\\
&+\mathbb{E}\left[\left(\int_0^T|\lambda(s)-\lambda_n(s)|ds\right)^2\right].
\end{aligned}$$

But since $\sup_{s\leq T}|p(s)-p_n(s)|\in[0,1]$, $\left(\sup_{s\leq T}|p(s)-p_n(s)|\right)^4\leq \sup_{s\leq T}|p(s)-p_n(s)|$ and hence

$$\begin{aligned}
\mathbb{E}[H_n^2]\leq &\left\{\mathbb{E}\left[\left(\sup_{s\leq T}|p(s)-p_n(s)|\right)\right]\mathbb{E}\left[\left(\int_0^T\lambda(s)ds\right)^4\right]\right\}^{\frac{1}{2}}\\
&+2\left\{\mathbb{E}\left[\left(\int_0^T\lambda(s)ds\right)^2\right]\mathbb{E}\left[\left(\int_0^T|\lambda(s)-\lambda_n(s)|ds\right)^2\right]\right\}^{\frac{1}{2}}\\
&+\mathbb{E}\left[\left(\int_0^T|\lambda(s)-\lambda_n(s)|ds\right)^2\right].
\end{aligned}$$

Note that the compensated Hawkes process $G(t) := M(t) - \int_0^t \lambda(s)ds$ is a martingale. Denote by $([G]_t)_{t\geq 0}$ as the (optional) quadratic variation process of G. Then by the Burkholder–Davis–Gundy inequality we have for all $\ell \in \mathbb{N}$ that

$$\mathbb{E}|G(T)|^\ell \leq \eta_\ell \mathbb{E}\left[([G]_T)^{\ell/2}\right] = \eta_\ell \mathbb{E}[(M(T))^{\ell/2}]$$

for some constant $\eta_\ell > 0$ that depends on ℓ. Then

$$\begin{aligned}\mathbb{E}\left[\left(\int_0^T \lambda(s)ds\right)^\ell\right] &= \mathbb{E}[|M(T) - G(T)|^\ell] \\ &\leq 2^{\ell-1}\left(\mathbb{E}[(M(T))^\ell] + \mathbb{E}[|G(t)|^\ell]\right) \\ &\leq 2^{\ell-1}\left(\mathbb{E}[(M(T))^\ell] + \eta_\ell \mathbb{E}[(M(T))^{\ell/2}]\right) \leq 2^{\ell-1}(A_{\Lambda,\rho,\ell}(T) + \eta_\ell A_{\Lambda,\rho,\ell/2}(T)),\end{aligned}$$

where $A_{\Lambda,\rho,\ell}(T)$ is the moment estimate of $\mathbb{E}[(M(T))^\ell]$ introduced in (A.5). In particular, $\mathbb{E}\left[\left(\int_0^T \lambda(s)ds\right)^2\right] < \infty$ and $\mathbb{E}\left[\left(\int_0^T \lambda(s)ds\right)^4\right] < \infty$. We conclude $\mathbb{E}[H_n^2] \to 0$ on noticing that

$$\mathbb{E}\left[\left(\int_0^T |\lambda(s) - \lambda_n(s)|ds\right)^2\right] \to 0, \qquad \mathbb{E}\left[\sup_{s\leq T}|p(s) - p_n(s)|\right] \to 0,$$

due to Corollary A.2.

Finally,

$$|Q(t) - Q_n(t)| = |(N^-(t) - N_n^-(t)) - (N^+(t) - N_n^+(t))|$$

such that

$$\begin{aligned}\mathbb{E}\left[\left(\sup_{0\leq u\leq T}|Q(t) - Q_n(t)|\right)^2\right] &\leq 2\mathbb{E}\left[\left(\sup_{0\leq u\leq T}|N^-(t) - N_n^-(t)|\right)^2\right] + 2\mathbb{E}\left[\left(\sup_{0\leq u\leq T}|N^+(t) - N_n^+(t)|\right)^2\right] \\ &\to 0.\end{aligned}$$

This concludes the proof. □

A.3 Almost-sure uniform convergence of the approximating state processes

The convergence (in expectation) of the approximating inventory processes established in Proposition A.3 relies on the assumption that there is no position limit imposed on Q. To handle the possibility of a finite inventory constraint (equivalently, $\underline{q} > -\infty$ or $\bar{q} < \infty$), a stronger condition (relative to Condition (A.20)) will help us establish a similar convergence result.

Proposition A.4. *If it holds that*

$$\min\left(\sum_{n=1}^\infty E_n^\pm(T), \sum_{n=1}^\infty \int_0^T \epsilon_n^\pm(t)dt\right) < \infty \qquad \textit{and} \qquad \lim_{n\to\infty} E_{\kappa,n}^\pm = 0, \tag{A.23}$$

then we have:

1. $\mathbb{P}\left(\liminf_{n\to\infty}\left\{\omega \in \Omega : \sup_{0\leq t\leq T}|M^\pm(t) - M_n^\pm(t)| = 0\right\}\right) = 1$;
2. $\mathbb{P}\left(\liminf_{n\to\infty}\left\{\omega \in \Omega : \sup_{0\leq t\leq T}|S(t) - S_n(t)| = 0\right\}\right) = 1$;
3. $\sup_{0\leq t\leq T}|\kappa^\pm(t) - \kappa_n^\pm(t)| \to 0$ $\mathbb{P}$-*a.s.*;

4. $\sup_{0\le t\le T}|p^{\pm}(t)-p_n^{\pm}(t)|\to 0$ $\mathbb{P}$-*a.s.;*

5. $\mathbb{P}\left(\liminf_{n\to\infty}\left\{\omega\in\Omega:\sup_{0\le t\le T}|N^{\pm}(t)-N_n^{\pm}(t)|=0\right\}\right)=1$;

6. $\mathbb{P}\left(\liminf_{n\to\infty}\left\{\omega\in\Omega:\sup_{0\le t\le T}|Q(t)-Q_n(t)|=0\right\}\right)=1$.

Proof. Using (A.6) in Proposition A.1 and Markov's inequality, we have for each n that

$$\mathbb{P}\left(\sup_{0\le t\le T}|M^{\pm}(t)-M_n^{\pm}(t)|>\alpha\right)\le\frac{\mathbb{E}[\sup_{0\le t\le T}|M^{\pm}(t)-M_n^{\pm}(t)|]}{\alpha}\le\frac{\Lambda}{\alpha(1-\rho)^2}\min\left(E_n^{\pm}(T),\int_0^T\epsilon_n^{\pm}(T)\right)$$

where we pick $\alpha\in(0,1)$. Hence

$$\sum_{n=1}^{\infty}\mathbb{P}\left(\sup_{0\le t\le T}|M^{\pm}(t)-M_n^{\pm}(t)|>\alpha\right)\le\frac{\Lambda}{\alpha(1-\rho)^2}\min\left(\sum_{n=1}^{\infty}E_n^{\pm}(T),\sum_{n=1}^{\infty}\int_0^T\epsilon_n^{\pm}(T)dt\right)<\infty.$$

By the first Borel–Cantelli lemma, we arrive at

$$\mathbb{P}\left(\liminf_{n\to\infty}\left\{\omega\in\Omega:\sup_{0\le t\le T}|M^{\pm}(t)-M_n^{\pm}(t)|\le\alpha\right\}\right)=1.$$

But $\sup_{0\le t\le T}|M^{\pm}(t)-M_n^{\pm}(t)|$ can only take values in the non-negative integers. Hence part (1) of the claim holds.

Next,

$$|S(t)-S_n(t)|=|\epsilon^{+}(M^{+}(t)-M_n^{+}(t))-\epsilon^{-}(M^{-}(t)-M_n^{-}(t))|$$

such that

$$\sup_{0\le t\le T}|S(t)-S_n(t)|\le\epsilon^{+}\sup_{0\le t\le T}|(M^{+}(t)-M_n^{+}(t))|+\epsilon^{-}\sup_{0\le t\le T}|(M^{-}(t)-M_n^{-}(t))|,$$

where by part (1) of this proposition there is probability of one that the right-hand-side of the above expression becomes exactly zero for sufficiently large n. Hence part (2) of the proposition holds.

By part (1) of the proposition, with probability one the processes M_n and M eventually become indistinguishable under large n. Hence, with probability one, there exists $\hat{n}=\hat{n}(\omega)$ such that we have $\tau_i^{\pm}=\tau_{i,n}^{\pm}$ for all $i\in\mathbb{N}$ and $n>\hat{n}$. In other words, the jump times of $M_n^{\pm}$ and $M^{\pm}$ must coincide under large n. Then we have

$$\begin{aligned}|\kappa^{\pm}(t)-\kappa_n^{\pm}(t)|&=\left|\int_0^t K_{\kappa}^{\pm}(t,s)dM^{\pm}(s)-\int_0^t K_{\kappa,n}^{\pm}(t,s)dM_n^{\pm}(s)\right|\\&=\left|\int_0^t K_{\kappa}^{\pm}(t,s)dM^{\pm}(s)-\int_0^t K_{\kappa,n}^{\pm}(t,s)dM^{\pm}(s)\right|\\&\le\int_0^t|K_{\kappa}^{\pm}(t,s)-K_{\kappa,n}^{\pm}(t,s)|dM^{\pm}(s)\\&\le M^{\pm}(t)E_{\kappa,n}^{\pm}\end{aligned}$$

and hence

$$\sup_{0\le t\le T}|\kappa^{\pm}(t)-\kappa_n^{\pm}(t)|\le M^{\pm}(T)E_{\kappa,n}^{\pm}.$$

The result follows from the fact that $M^{\pm}(T)<\infty$ almost surely and Condition (A.23). Now part (4) immediately follows from the fundamental inequality $|e^{-x}-e^{-y}|\le|x-y|$ for $x,y\ge 0$.

It remains to prove part (5) and (6) of the proposition. We assume without loss of generality that $q_0 = 0$. Write $\mathcal{T}^+ := \{\tau_i^+ : i \in \mathbb{N}, \tau_i^+ \leq T\}$ and $\mathcal{T}^- := \{\tau_i^- : i \in \mathbb{N}, \tau_i \leq T\}$, and define $\mathcal{S}$ as the ordered set of the union $\mathcal{T}^+ \cup \mathcal{T}^-$. Let η_i be the ith jump time of Q. Since $Q_n(t) := M_n^-(t) - M_n^+(t)$ and $Q(t) := M^-(t) - M^+(t)$, for large n the jump times of Q_n and Q must coincide as well and are given by each $\eta_i \in \mathcal{S}$.

Since $N_n^\pm$, Q_n, $N^\pm$ and Q have common initial values of zero at $t = 0$, it is sufficiently to show for each i that

$$|N_n^\pm(\eta_i) - N^\pm(\eta_i)| \overset{\text{a.s.}}{\to} 0, \qquad |Q_n(\eta_i) - Q(\eta_i)| \overset{\text{a.s.}}{\to} 0, \tag{A.24}$$

as $n \to \infty$. Then using the facts that these processes take values of integers, (A.24) would then imply $|N_n^\pm(\eta_i) - N^\pm(\eta_i)| = 0$ and $|Q_n(\eta_i) - Q(\eta_i)| = 0$ eventually for large n and this in turn would establish part (5) and (6) of this proposition. In what follows, we prove (A.24) by induction.

Suppose $\eta_1 \in \mathcal{T}^+$. Then $\eta_1 = \tau_1^+ < \tau_1^-$. As N_n^- is a thinned version of M_n^-, we have $N_n^-(t) \leq M_n^-(t)$ for all t and n. Then for large n, $N_n^-(\eta_1) \leq M_n^-(\eta_1) = M^-(\eta_1) \leq M^-(\tau_1^- -) = 0$ and thus $N_n^-(\eta_1) = 0$. Likewise, we can establish $N^-(\eta_1) = 0$ such that $|N_n^-(\eta_1) - N^-(\eta_1)| = 0$. Observe that $Q(\tau_1^+ -) = Q(\eta_1 -) = 0 = Q_n(\eta_1 -) = Q_n(\tau_1^+ -)$ for all n. Then using the representation in (2.3), we have on the set $\{\omega \in \Omega : \eta_1 \in \mathcal{T}^+\}$ that

$$\begin{aligned}
|N_n^+(\eta_1) - N^+(\eta_1)| &= |N_n^+(\tau_1^+) - N^+(\tau_1^+)| \\
&= |\mathbb{1}_{\{U_1^+ < p_n^+(\tau_1^+ -), Q_n(\tau_1^+ -) > \underline{q}\}} - \mathbb{1}_{\{U_1^+ < p^+(\tau_1^+ -), Q(\tau_1^+ -) > \underline{q}\}}| \\
&= |\mathbb{1}_{\{U_1^+ < p_n^+(\tau_1^+ -)\}} - \mathbb{1}_{\{U_1^+ < p^+(\tau_1^+ -)\}}| \times \mathbb{1}_{\{Q(\tau_1^+ -) > \underline{q}\}},
\end{aligned}$$

where U_1^+ is some uniform random variable independent of $\mathcal{F}_{\tau_1^+ -}$. Using the result in part (4) that p_n converges to p uniformly and the fact that U_1^+ is a uniform random variable such that the $\mathbb{P}(U_1^+ = p^+(\tau_1^+ -)) = 0$, we deduce $|\mathbb{1}_{\{U_1^+ < p_n^+(\tau_1^+ -)\}} - \mathbb{1}_{\{U_1^+ < p^+(\tau_1^+ -)\}}| \to 0$ almost surely and hence $|N_n^+(\eta_1) - N^+(\eta_1)| \to 0$ on the set $\{\omega \in \Omega : \eta_1 \in \mathcal{T}^+\}$. The alternative case of $\eta_1 \in \mathcal{T}^-$ can be handled similarly to conclude $|N_n^\pm(\eta_1) - N^\pm(\eta_1)| \to 0$ almost surely. This then implies $|Q_n(\eta_1) - Q(\eta_1)| \to 0$ almost surely. Hence (A.24) holds for $i = 1$.

Assume there exists some $k \in \mathbb{N}$ such that (A.24) holds for $i = 1, 2, ..., k$. The almost-sure convergence of integer-valued random variables implies eventual convergence, i.e. with probability one there exists large enough n such that $N_n^\pm(\eta_i) = N^\pm(\eta_i)$ and $Q_n(\eta_i) = Q(\eta_i)$ for all $i = 1, ..., k$.

We can again distinguish two cases. Suppose $\eta_{k+1} \in \mathcal{T}^+$. Then $N_n^-(t)$ and $N^-(t)$ do not jump over the time interval $t \in (\eta_k, \eta_{k+1}]$ such that $N_n^-(\eta_{k+1}) = N_n^-(\eta_k)$ and $N^-(\eta_{k+1}) = N^-(\eta_k)$. But $N_n^-(\eta_k) = N^-(\eta_k)$ for sufficiently large n by the induction hypothesis. Hence $|N_n^-(\eta_{k+1}) - N^-(\eta_{k+1})| = 0$ for large n. Moreover, on the set $\{\omega \in \Omega : \eta_{k+1} \in \mathcal{T}^+\}$ we have

$$\begin{aligned}
|N_n^+(\eta_{k+1}) - N^+(\eta_{k+1})| &= |\mathbb{1}_{\{U_{k+1}^+ < p_n^+(\eta_{k+1} -), Q_n(\eta_{k+1} -) > \underline{q}\}} - \mathbb{1}_{\{U_{k+1}^+ < p^+(\eta_{k+1} -), Q(\eta_{k+1} -) > \underline{q}\}}| \\
&= |\mathbb{1}_{\{U_{k+1}^+ < p_n^+(\eta_{k+1} -)\}} - \mathbb{1}_{\{U_{k+1}^+ < p^+(\eta_{k+1} -)\}}| \times \mathbb{1}_{\{Q(\eta_{k+1} -) > \underline{q}\}} \to 0
\end{aligned}$$

almost surely, where we have used the induction hypothesis $Q_n(\eta_k) = Q(\eta_k)$ for large n such that $Q_n(\eta_{k+1} -) = Q_n(\eta_k) = Q(\eta_k) = Q(\eta_{k+1} -)$, and the observation that

$$|\mathbb{1}_{\{U_{k+1}^+ < p_n^+(\eta_{k+1} -)\}} - \mathbb{1}_{\{U_{k+1}^+ < p^+(\eta_{k+1} -)\}}| \to 0$$

almost surely based on the same arguments for the case of $i = 1$. The other possible case of $\eta_{k+1} \in \mathcal{T}^-$ can again be handled similarly for us to deduce $|N_n^\pm(\eta_{k+1}) - N^\pm(\eta_{k+1})| \to 0$ and in turn $|Q_n(\eta_{k+1}) - Q(\eta_{k+1})| \to 0$ almost surely. Hence (A.24) holds for $i = k + 1$ too, and it therefore holds for all $i \in \mathbb{N}$ by the principle of induction. □

Proposition A.5. *Suppose* (A.23) *holds. Then*

$$\lim_{n\to\infty}\mathbb{E}\left[\left(\sup_{t\le T}|N^{\pm}(t)-N_n^{\pm}(t)|\right)^2\right]=\lim_{n\to\infty}\mathbb{E}\left[\left(\sup_{t\le T}|Q(t)-Q_n(t)|\right)^2\right]=0.$$

Proof. We verify the result for N_n^+, where identical arguments can be used to establish the result for N_n^-. Following similar derivation in the proof of Proposition A.3, we can deduce

$$\mathbb{E}\left[\left(\sup_{t\le T}|N^{+}(t)-N_n^{+}(t)|\right)^2\right]\le\mathbb{E}[H_n]+\mathbb{E}[H_n^2]. \tag{A.25}$$

where

$$H_n:=\int_0^T|\lambda^+(s-)p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-\lambda_n^+(s-)p_n^+(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}|ds.$$

As in Proposition A.3, we just need to show that $\mathbb{E}[H_n^2]\to 0$. Now,

$$\begin{aligned}
H_n&=\int_0^T|\lambda^+(s-)p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-\lambda_n^+(s-)p_n^+(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}|ds\\
&\le\int_0^T\lambda^+(s-)|p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-p_n(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}|ds\\
&\quad+\int_0^T p_n^+(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}|\lambda^+(s-)-\lambda_n^+(s-)|ds\\
&\le\int_0^T\lambda^+(s-)|p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-p_n(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}|ds+\int_0^T|\lambda^+(s-)-\lambda_n^+(s-)|ds\\
&\le\sup_{s\le T}\left|p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-p_n(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}\right|\int_0^T\lambda^+(s)ds+\int_0^T|\lambda^+(s)-\lambda_n^+(s)|ds.
\end{aligned}$$

Taking square on the both sides gives

$$\begin{aligned}
H_n^2\le&\left(\sup_{s\le T}\left|p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-p_n(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}\right|\right)^2\left(\int_0^T\lambda^+(s)ds\right)^2\\
&+2\left(\int_0^T\lambda^+(s)ds\right)\left(\int_0^T|\lambda^+(s)-\lambda_n^+(s)|ds\right)\\
&+\left(\int_0^T|\lambda^+(s-)-\lambda_n^+(s)|ds\right)^2.
\end{aligned}$$

Taking expectation and using the Cauchy-Schwarz inequality, we deduce

$$\begin{aligned}
\mathbb{E}[H_n^2]\le&\,\mathbb{E}\left[\left(\sup_{s\le T}\left|p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-p_n(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}\right|\right)^2\left(\int_0^T\lambda^+(s)ds\right)^2\right]\\
&+2\left\{\mathbb{E}\left[\left(\int_0^T\lambda^+(s)ds\right)^2\right]\mathbb{E}\left[\left(\int_0^T|\lambda^+(s)-\lambda_n^+(s)|ds\right)^2\right]\right\}^{\frac{1}{2}}\\
&+\mathbb{E}\left[\left(\int_0^T|\lambda^+(s)-\lambda_n^+(s)|ds\right)^2\right].
\end{aligned}$$

The second and the third term on the right-hand-side of the above inequality vanish as $n\to\infty$. For the first term, we have

$$\left(\sup_{s\le T}\left|p(s-)\mathbb{1}_{\{Q(s-)>\underline{q}\}}-p_n(s-)\mathbb{1}_{\{Q_n(s-)>\underline{q}\}}\right|\right)^2\to 0$$

almost surely by Proposition A.4 and the random variable inside the first expectation is bounded from the above by $\left(\int_0^T \lambda^+(s)ds\right)^2$, which expectation is finite. Hence Dominated Convergence Theorem implies the first expectation converges to zero as well, and subsequently $\mathbb{E}\left[H_n^2\right] \to 0$. Finally,

$$\begin{aligned}\mathbb{E}\left[\left(\sup_{0\le u\le t}|Q(t)-Q_n(t)|\right)^2\right] &\le 2\mathbb{E}\left[\left(\sup_{0\le u\le t}|N^-(t)-N_n^-(t)|\right)^2\right] + 2\mathbb{E}\left[\left(\sup_{0\le u\le t}|N^+(t)-N_n^+(t)|\right)^2\right]\\ &\to 0.\end{aligned}$$

The result immediately follows. □

It is worth mentioning that the results in Proposition A.5 will hold for as long as Condition (A.23) is in place, regardless of whether $\min(|\underline{q}|, \bar{q})$ is finite or not.

A.4 Convergence of the approximating value functions

Following the notations introduced in Section A.3, define

$$\begin{aligned}J = J(\delta) &:= \mathbb{E}\left[X(T) + Q(T)S(T) - \alpha Q^2(T) - \beta\int_0^T Q^2(u)du\right],\\ J_n = J_n(\delta) &:= \mathbb{E}\left[X_n(T) + Q_n(T)S_n(T) - \alpha Q_n^2(T) - \beta\int_0^T Q_n^2(u)du\right],\end{aligned}$$

which represent two copies of the objective functions driven by Y and Y_n defined on the same probability space. These are evaluated against a common admissible strategy $\delta = (\delta_t^+, \delta_t^-)_{t\ge 0} \in \mathcal{A}$.

Proposition A.6. *Suppose we are in either one of the following cases:*

1. *Condition* (A.23) *holds, or*
2. *Condition* (A.20) *holds and there is no inventory constraint, i.e.* $\underline{q} = -\infty$ *and* $\hat{q} = +\infty$.

Then $|J(\delta) - J_n(\delta)| \to 0$ *for any admissible* $\delta \in \mathcal{A}$ *as* $n \to \infty$.

Proof. We first show that $\mathbb{E}|X(T) - X_n(T)| \to 0$. Observe that

$$\begin{aligned}&\left|\int_0^t (S(u-)+\delta^+(u))dN^+(u) - \int_0^t (S_n(u-)+\delta^+(u))dN_n^+(u)\right|\\ &\le \int_0^t |(S(u-)-S_n(u-))|dN_n^+(u) + \int_0^t |(S(u-)+\delta^+(u))|d|N^+ - N_n^+|(u)\\ &\le \left(\sup_{u\le t}|S(u-)-S_n(u-)|\right)N_n^+(t) + \left(C + \sup_{u\le t}|S(u-)|\right)|N^+(t) - N_n^+(t)|\end{aligned}$$

and similarly

$$\begin{aligned}&\left|\int_0^t (S(u-)-\delta^-(u))dN^-(u) - \int_0^t (S_n(u-)-\delta^-(u))dN_n^-(u)\right|\\ &\le \left(\sup_{u\le t}|S(u-)-S_n(u-)|\right)N_n^-(t) + \left(C + \sup_{u\le t}|S(u-)|\right)|N^-(t) - N_n^-(t)|\end{aligned}$$

Then we have

$$
\begin{aligned}
\mathbb{E}|X(T)-X_n(T)| &= \mathbb{E}\Bigg|\int_0^T (S(u-)+\delta^+(u))dN^+(u) - \int_0^T (S(u-)-\delta^-(u))dN^-(u) \\
&\quad - \int_0^T (S_n(u-)+\delta^+(u))dN_n^+(u) + \int_0^T (S_n(u-)-\delta^-(u))dN_n^-(u)\Bigg| \\
&\le \mathbb{E}\left[\left(\sup_{u\le T}|S(u-)-S_n(u-)|\right)N_n^+(T)\right] + \mathbb{E}\left[\left(C+\sup_{u\le T}|S(u-)|\right)|N^+(T)-N_n^+(T)|\right] \\
&\quad + \mathbb{E}\left[\left(\sup_{u\le T}|S(u-)-S_n(u-)|\right)N_n^-(T)\right] + \mathbb{E}\left[\left(C+\sup_{u\le T}|S(u-)|\right)|N^-(T)-N_n^-(T)|\right] \\
&\le \left(\mathbb{E}\left[\left(\sup_{u\le T}|S(u-)-S_n(u-)|\right)^2\right]\right)^{\frac{1}{2}}\left(\mathbb{E}\left[\left(N_n^+(t)\right)^2\right]\right)^{\frac{1}{2}} \\
&\quad + \left(\mathbb{E}\left[\left(C+\sup_{u\le T}|S(u-)|\right)^2\right]\right)^{\frac{1}{2}}\left(\mathbb{E}\left[(N^+(T)-N_n^+(T))^2\right]\right)^{\frac{1}{2}} \\
&\quad + \left(\mathbb{E}\left[\left(\sup_{u\le T}|S(u-)-S_n(u-)|\right)^2\right]\right)^{\frac{1}{2}}\left(\mathbb{E}\left[\left(N_n^-(t)\right)^2\right]\right)^{\frac{1}{2}} \\
&\quad + \left(\mathbb{E}\left[\left(C+\sup_{u\le T}|S(u-)|\right)^2\right]\right)^{\frac{1}{2}}\left(\mathbb{E}\left[(N^-(T)-N_n^-(T))^2\right]\right)^{\frac{1}{2}},
\end{aligned}
$$

where we have used the Cauchy-Schwarz inequality. It is trivial that $\mathbb{E}\left[\left(N_n^{\pm}(T)\right)^2\right] \le \mathbb{E}\left[\left(M_n^{\pm}(T)\right)^2\right] \le A_{\Lambda,\rho,2}(T) < \infty$ for all n since $N_n^{\pm}$ are just the thinned versions of $M_n^{\pm}$, and it is also clear that $\mathbb{E}\left[\left(\sup_{u\le T}|S(u-)|\right)^2\right] < \infty$ because S is a linear combination of a Brownian motion and two Hawkes processes. The right-hand-side of the above inequality therefore converges to zero in both Case 1 and Case 2 due to part (2) of Corollary A.2, Proposition A.5 (for Case 1) and Proposition A.3 (for Case 2).

We therefore arrive at

$$
\begin{aligned}
|J-J_n| &\le \mathbb{E}|X(T)-X_n(T)| + \mathbb{E}|Q_n(T)(S(T)-S_n(T))| + \mathbb{E}|S(T)(Q(T)-Q_n(T))| \\
&\quad + \alpha\mathbb{E}|Q^2(T)-Q_n^2(T)| + \beta\int_0^T \mathbb{E}|Q^2(u)-Q_n^2(u)|du \\
&\le \mathbb{E}|X(T)-X_n(T)| + \mathbb{E}|Q_n(T)(S(T)-S_n(T))| + \mathbb{E}|S(T)(Q(T)-Q_n(T))| \\
&\quad + \alpha\mathbb{E}(|Q(T)+Q_n(T)|\cdot|Q(T)-Q_n(T)|) \\
&\quad + \beta T\mathbb{E}\left[\left(\sup_{0\le u\le T}|Q_n(u)+Q(u)|\right)\left(\sup_{0\le u\le T}|Q_n(u)-Q(u)|\right)\right] \\
&\le \mathbb{E}|X(T)-X_n(T)| + \left(\mathbb{E}[(Q_n(T))^2]\right)^{\frac{1}{2}}\left(\mathbb{E}[(S(T)-S_n(T))^2]\right)^{\frac{1}{2}} + \left(\mathbb{E}[S^2(T)]\right)^{\frac{1}{2}}\left(\mathbb{E}[(Q(T)-Q_n(T))^2]\right)^{\frac{1}{2}} \\
&\quad + \alpha\left(\mathbb{E}[(|Q_n(T)+Q(T)|)^2]\right)^{\frac{1}{2}}\left(\mathbb{E}[(Q(T)-Q_n(T))^2]\right)^{\frac{1}{2}} \\
&\quad + \beta T\left[\mathbb{E}\left[\left(\sup_{0\le u\le T}|Q(u)+Q_n(u)|\right)^2\right]\right]^{\frac{1}{2}}\left[\mathbb{E}\left[\left(\sup_{0\le u\le T}|Q(u)-Q_n(u)|\right)^2\right]\right]^{\frac{1}{2}},
\end{aligned}
$$

where we have used the Cauchy-Schwarz inequality again. It is not hard to see $\mathbb{E}[(Q_n(T))^2]$ and $\mathbb{E}\left[\left(\sup_{0\le u\le T}|Q(u)+Q_n(u)|\right)^2\right]$ are bounded for all n due to the fact that $\mathbb{E}\left[\left(M_n^{\pm}(T)\right)^2\right] \le A_{\Lambda,\rho,2}(T) < \infty$. We conclude $|J-J_n| \to 0$ using the previously shown convergence result of $\mathbb{E}|X(T)-X_n(T)| \to 0$, part (2) of Corollary A.2, Proposition A.5 (for Case 1) and Proposition A.3 (for Case 2) □

We are now finally ready to prove Theorem 2.4.

Proof of Theorem 2.4. Recall that we work under the assumption of $a_{\lambda^\pm} = a_{\kappa^\pm} = 1$ without loss of generality. For $K^\pm$ and $K^\pm_\kappa$ which satisfy the conditions in Theorem 2.3, define

$$K^\pm_n(t,s) := \sum_{i=1}^n \xi^\pm_n \varphi_n(t)\varphi^\pm_n(s), \qquad K^\pm_{\kappa,n}(t,s) := \sum_{i=1}^n \xi^\pm_{\kappa,n}\varphi_{\kappa,n}(t)\varphi^\pm_{\kappa,n}(s),$$

as the truncated Mercer's expansion of $K^\pm$ and $K^\pm_\kappa$ respectively. We first show that there exists a constant $\rho \in (0,1)$ and $\hat{n} \in \mathbb{N}$ such that $(K^+_n, K^-_n, K^+_{\kappa,n}, K^-_{\kappa,n}) \in \mathcal{H}_\rho$ for all $n > \hat{n}$ where $\mathcal{H}_\rho$ is defined in (A.1).

We now drop the superscript $\pm$ for brevity. Since K_n converges to K uniformly on $[0,T]^2$ by Theorem 2.3, we have

$$\sup_{t\in[0,T]} \int_0^t |K_n(t,s) - K(t,s)|ds \le T \sup_{(t,s)\in[0,T]^2} |K_n(t,s) - K(t,s)| \to 0.$$

In particular, for any $\epsilon_1 > 0$ there exists $\hat{n}_1 = \hat{n}_1(\epsilon_1)$ such that

$$\sup_{t\in[0,T]} \int_0^t |K_n(t,s) - K(t,s)|ds < \epsilon_1$$

for all $n > \hat{n}_1$, and in turn

$$\begin{aligned}\sup_{t\in[0,T]} \int_0^t |K_n(t,s)|ds &< \sup_{t\in[0,T]} \int_0^t |K(t,s)|ds + \epsilon_1 \\ &= \sup_{t\in[0,T]} \int_0^t g(|t-s|)ds + \epsilon_1 \\ &\le ||g||_1 + \epsilon_1.\end{aligned}$$

Likewise, we can deduce for any $\epsilon_2 > 0$ there exists $\hat{n}_2 = \hat{n}_2(\epsilon_2)$ such that

$$\sup_{s\in[0,T]} \int_s^T |K_n(t,s)|dt < ||g||_1 + \epsilon_2.$$

Hence if we pick some $\rho \in (||g||_1, 1)$ and set $\epsilon_1 = \epsilon_2 = \rho - ||g||_1 > 0$, we get for $n > \max(\hat{n}_1, \hat{n}_2)$ that

$$\max\left(\sup_{t\in[0,T]} \int_0^t |K_n(t,s)|ds, \sup_{s\in[0,T]} \int_s^T |K_n(t,s)|dt\right) < \rho,$$

i.e. $K_n \in \mathcal{H}_\rho$. This line of arguments can be easily extended to show the existence of some common $\rho \in (0,1)$ and $\hat{n} \in \mathbb{N}$ such that $(K^+_n, K^-_n, K^+_{\kappa,n}, K^-_{\kappa,n}) \in \mathcal{H}_\rho$ for all $n > \hat{n}$.

Observe that

$$E^\pm_{\kappa,n} := \sup_{(t,s)\in\mathcal{D}} |K^\pm_\kappa(t,s) - K^\pm_{\kappa,n}(t,s)| \le \sup_{(t,s)\in[0,T]^2} |K^\pm_\kappa(t,s) - K^\pm_{\kappa,n}(t,s)| \to 0$$

by the uniform convergence of $K^\pm_{\kappa,n}$ to $K^\pm_\kappa$. Likewise,

$$E^\pm_n(T) := \sup_{u\in[0,T]} \int_u^T |K^\pm(s,u) - K^\pm_n(s,u)|ds \le T \sup_{(s,u)\in[0,T]^2} |K^\pm(s,u) - K^\pm_n(s,u)| \to 0.$$

Hence Condition (A.20) is satisfied by $K^\pm_n$ and $K^\pm_{\kappa,n}$.

Next, we show that Condition (A.23) is satisfied under the further assumption (2.12). Check that

$$\begin{aligned}
\epsilon_n^{\pm}(t) &= \int_0^T |K^{\pm}(t,u) - K_n^{\pm}(t,u)|du \\
&= \int_0^T \left| \sum_{i=n+1}^{\infty} \xi_i^{\pm} \varphi_i^{\pm}(t) \varphi_i^{\pm}(u) \right| du \\
&\leq \sqrt{T} \left[\int_0^T \left(\sum_{i=n+1}^{\infty} \xi_i^{\pm} \varphi_i^{\pm}(t) \varphi_i^{\pm}(u) \right)^2 ds \right]^{\frac{1}{2}} \\
&= \sqrt{T} \left[\sum_{i=n+1}^{\infty} (\xi_i^{\pm})^2 (\varphi_i^{\pm}(t))^2 \right]^{\frac{1}{2}},
\end{aligned}$$

where we have used the Cauchy-Schwarz inequality and the fact that $\{\varphi_j^{\pm}\}_{j\in\mathbb{N}}$ are orthonormal on $L^2([0,T])$. Then

$$\begin{aligned}
\int_0^T \epsilon_n^{\pm}(t)dt &= \sqrt{T} \int_0^T \left[\sum_{i=n+1}^{\infty} (\xi_i^{\pm})^2 (\varphi_i^{\pm}(t))^2 \right]^{\frac{1}{2}} dt \\
&\leq T \left[\int_0^T \sum_{i=n+1}^{\infty} (\xi_i^{\pm})^2 (\varphi_i^{\pm}(t))^2 dt \right]^{\frac{1}{2}} = T \left[\sum_{i=n+1}^{\infty} (\xi_i^{\pm})^2 \right]^{\frac{1}{2}}.
\end{aligned}$$

on using one more time the Cauchy-Schwarz inequality and the orthonormality of $\{\varphi_j^{\pm}\}_{j\in\mathbb{N}}$. Hence

$$\sum_{n=1}^{\infty} \int_0^T \epsilon_n^{\pm}(t)dt < \infty$$

and in turn Condition (A.23) holds provided that $\sum_{n=1}^{\infty} \left[\sum_{i=n+1}^{\infty} (\xi_i^{\pm})^2\right]^{\frac{1}{2}} < \infty$.

We have verified that, for large enough n, the sequence of kernels $(K_n^+, K_n^-, K_{\kappa,n}^+, K_{\kappa,n}^-)$ belongs to the class $\mathcal{H}_\rho^4$ and falls into either Case 1 or Case 2 of Proposition A.6. Then we have

$$|V - V_n| = \left| \sup_{\delta\in\mathcal{A}} J(\delta) - \sup_{\delta\in\mathcal{A}} J_n(\delta) \right| \leq \sup_{\delta\in\mathcal{A}} |J(\delta) - J_n(\delta)| \to 0.$$

□